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Dietary weight loss-induced improvements in metabolic function are enhanced by exercise in people with obesity and prediabetes

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The additional therapeutic effects of regular exercise during a dietary weight loss program in people with obesity and prediabetes are unclear. Here, we show that whole-body (primarily muscle) insulin sensitivity (primary outcome) was 2-fold greater ($P = 0.006$) after 10% weight loss induced by calorie restriction plus exercise training (Diet+EX; $n = 8$, 6 women) than 10% weight loss induced by calorie restriction alone (Diet-ONLY; $n = 8$, 4 women) in participants in two concurrent studies. The greater improvement in insulin sensitivity was accompanied by increased muscle expression of genes involved in mitochondrial biogenesis, energy metabolism and angiogenesis (secondary outcomes) in the Diet+EX group. There were no differences between groups in plasma branched-chain amino acids or markers of inflammation, and both interventions caused similar changes in the gut microbiome. Few adverse events were reported. These results demonstrate that regular exercise during a diet-induced weight loss program has profound additional metabolic benefits in people with obesity and prediabetes.

Trial Registration: ClinicalTrials.gov (NCT02706262 and NCT02706288)

Insulin-resistant glucose metabolism is the most common metabolic abnormality associated with obesity and is involved in the pathogenesis of obesity-related cardiometabolic comorbidities, including dyslipidemia, nonalcoholic fatty liver disease (NAFLD), type 2 diabetes and cardiovascular disease[1]. Diet-induced weight loss and exercise training without weight loss improve insulin sensitivity and the metabolic abnormalities associated with obesity[2–6]. Therefore, lifestyle interventions that combine an energy-deficit diet and increased physical activity are recommended as the primary therapy for people with obesity[7–9]. However, the additional benefit of exercise training, when provided in conjunction with diet-induced weight loss, on insulin sensitivity and metabolic health is unclear because of conflicting data from previous

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studies, which have reported an additive effect[10–14] or no additional effect of exercise training on insulin sensitivity[15–19]. The reason(s) for the discrepant findings among studies could be related to differences in the methods used to assess insulin sensitivity or the duration between the most recent exercise bout and metabolic testing. A better understanding of the potential additional therapeutic effects of exercise training beyond diet-induced weight loss alone is important because exercise programs are often more difficult to implement than calorie restriction in people with obesity owing to increased barriers to exercise in this population[20].

The main purpose of the present study was to test the hypothesis that moderate diet-induced weight loss (10%) in conjunction with multimodal exercise training causes greater improvement in whole-body (primarily muscle[21]) insulin sensitivity (primary outcome) than matched 10% weight loss induced by calorie restriction alone in people with obesity and prediabetes. In addition, we assessed several secondary outcomes to provide a better understanding of whether weight loss with exercise training affects a constellation of other key therapeutic cardiometabolic outcomes and specific putative mechanisms proposed to regulate insulin action compared with weight loss alone. To this end, we conducted a comprehensive evaluation of the following outcomes in people with obesity and prediabetes before and after 10% weight loss induced within ~5 months by either calorie restriction alone (Diet-ONLY) or calorie restriction plus supervised exercise training (Diet+EX): (1) whole-body (muscle) and liver insulin sensitivity, assessed using the hyperinsulinemic-euglycemic clamp procedure in conjunction with stable isotopically labeled glucose tracer infusion; (2) β-cell function and insulin kinetics; (3) metabolic response to glucose ingestion assessed for 3 h and ingestion of multiple mixed meals assessed over 24 h; (4) cardiometabolic risk factors (plasma lipid profile, markers of inflammation, plasma branched-chain amino acid (BCAA) concentrations, body composition (fat-free mass (FFM), fat mass and intrahepatic triglyceride (IHTG) content), cardiorespiratory fitness and muscle strength); (5) skeletal muscle tissue global transcriptomics; (6) the plasma proteome; and (7) the composition of the gut microbiome. The assessment of the secondary outcomes was considered "hypothesis generating," so the statistical analysis of differences between groups was not corrected for multiple testing. The intervention in the Diet-ONLY group involved weekly individual dietary and behavioral education sessions, with all food provided as packed-out meals to enhance compliance and minimize the potential confounding effect of differences in macronutrient intake between groups on our study outcomes. The Diet+EX group had the same dietary intervention as the Diet-ONLY group plus 6 h of exercise training per week (four 1-h supervised exercise sessions and two 1-h sessions performed at home). A plant-forward diet, which is low in fat, sodium, and refined carbohydrates and high in complex carbohydrates, was used as the dietary intervention because this diet enhances weight loss and has considerable beneficial effects on glycemic control and other cardiometabolic abnormalities in people with obesity[22–24].

Results

Muscle strength and cardiorespiratory fitness

Mean weight loss was nearly identical in both groups (10.4 ± 0.9% and 10.6 ± 1.1% weight loss in the Diet-ONLY and Diet+EX groups, respectively) and was associated with decreases in body fat, FFM, appendicular lean mass and IHTG content that were not different between groups (Table 1). The duration of the intervention in the Diet-ONLY group (20.6 ± 1.8 weeks) was not different than the duration of the intervention in the Diet+EX group (18.2 ± 1.3 weeks) ($P = 0.31$). Participants in the Diet+EX group attended 97 ± 3% of the supervised exercise sessions. Muscle strength (assessed as the total weight (in kilograms) lifted during leg press, knee flexion, chest press and seated row one-repetition maximum (1-RM) strength tests) increased more in the Diet+EX group (13 ± 4% change from baseline) than in the Diet-ONLY group (−2 ± 3% change from baseline) ($P = 0.025$) (Table 1 and Extended Data Table 1). In addition, cardiorespiratory fitness (assessed as peak oxygen consumption (VO_{2peak}) during incremental treadmill exercise to volitional exhaustion) increased more in the Diet+EX group than the Diet-ONLY group when the data were expressed as the total rate of oxygen consumption (in L/min) (+10 ± 4% versus −6 ± 1%, respectively; $P = 0.007$), the rate of oxygen consumption in relation to FFM (in mL/kg FFM/min) (+13 ± 4% versus −3 ± 1%, respectively; $P = 0.005$), and the rate of oxygen consumption in relation to body weight (in mL/kg/min) (+23 ± 4% versus +6 ± 1%, respectively; $P = 0.004$) (Table 1 and Extended Data Table 1).

Hemoglobin A1c (HbA1c), fasting plasma glucose, triglycerides, BCAAs, high-density lipoprotein (HDL) cholesterol, total cholesterol and plasminogen activator inhibitor 1 (PAI-1) concentration decreased or tended to decrease after weight loss in both groups, without a difference between groups (Table 1 and Extended Data Table 2). Fasting plasma insulin decreased more after weight loss in Diet+EX group than in the Diet-ONLY group ($P = 0.009$) (Table 1). Fasting plasma nonesterified fatty acid (NEFA) concentrations were unchanged in the Diet-ONLY group but increased by ~50% in the Diet+EX group (difference between groups, $P = 0.022$). Weight loss did not change plasma low-density lipoprotein (LDL) cholesterol, C-reactive protein concentration or concentrations of many proinflammatory (MCP-1, IL-6, TNF-α, IL-17, IL-1β and IFNγ) or anti-inflammatory (IL-10) cytokines (Extended Data Table 2).

Metabolic response to glucose and mixed-meal ingestion

Plasma glucose concentrations for 3 h after participants ingested 75 g of glucose and for 24 h with the ingestion of three mixed meals (each providing one-third of the participant's estimated energy requirement[25], given at 0700 hours ($t = 0$ h), 1300 hours ($t = 6$ h) and 1900 hours ($t = 12$ h); Extended Data Fig. 1) decreased after weight loss in both groups, without a difference in the decline in plasma glucose areas under the curve (AUCs) between groups (Figs. 1a and 2a and Extended Data Table 3). Plasma insulin concentration AUCs for 3 h after glucose ingestion and for 24 h (with three mixed meals) decreased after weight loss in both groups, but the decreases were greater in the Diet+EX group than in the Diet-ONLY group ($P = 0.008$ and $P = 0.030$, respectively) (Figs. 1b and 2b and Extended Data Table 3). The decrease in postprandial insulin concentration in the Diet-ONLY group was due to a decrease in insulin secretion rate (ISR), whereas the greater decrease in postprandial insulin in the Diet+EX group was due to both a decrease in ISR and an increase in insulin clearance rate (ICR) (Figs. 1c, d and 2c, d and Extended Data Table 3). β-cell function (assessed as the relationship between ISR and plasma glucose concentration during the first 30 min after glucose ingestion when plasma glucose concentration is rapidly rising) was not different after weight loss in either the Diet+EX group or the Diet-ONLY group (Fig. 3a). In contrast, ICR at any given plasma insulin concentration was greater after weight loss in the Diet+EX group but did not change in the Diet-ONLY group (Fig. 3b). Plasma NEFA 24-h AUC decreased after weight loss in the Diet-ONLY group but increased after weight loss in the Diet+EX group, so the change in NEFA 24-h AUC was significantly different between groups ($P = 0.003$) (Fig. 2e and Extended Data Table 3).

Insulin sensitivity

Whole-body insulin sensitivity (assessed as the glucose disposal rate per kilogram FFM divided by plasma insulin concentration during a hyperinsulinemic-euglycemic clamp procedure, which primarily represents skeletal muscle insulin sensitivity[21]) increased after weight loss in both groups, but the increase was ~3-fold greater in the Diet+EX group than in the Diet-ONLY group ($P = 0.006$) (Fig. 4a and Extended Data Table 3). Hepatic insulin sensitivity (assessed as the reciprocal of the product of basal endogenous glucose production rate and basal plasma insulin concentration[26]) increased after weight loss in both groups, but the increase was ~2-fold greater in the Diet+EX group than

Table 1 | Participant characteristics before and after intervention

	Diet-ONLY (*n*=8)			Diet+EX (*n*=8)			Between groups
	Before	After	Difference (95% CI)	Before	After	Difference (95% CI)	*P* value
Body composition							
Body mass (kg)	114.8±8.7	103.3±8.7	−11.5 (−13.3, −9.7)	116.5±10.6	104.2±9.6	−12.3 (−16.3, −8.3)	0.682
BMI (kg/m^2)	39.1±1.9	35.2±2.1	−3.9 (−4.5, −3.4)	39.6±2.2	35.4±2.0	−4.2 (−5.3, −3.1)	0.676
Body fat (%)	45.6±2.6	41.3±3.2	−4.3 (−5.8, −2.8)	47.3±1.2	42.9±1.0	−4.4 (−5.5, −3.2)	0.981
FFM (kg)	61.1±4.4	59.1±4.4	−1.9 (−2.9, −1.0)	60.6±5.4	59.1±5.1	−1.5 (−2.6, −0.3)	0.502
Appendicular lean mass (kg)	27.8±2.2	26.3±2.1	−1.5 (−2.1, −0.9)	28.8±3.2	27.6±2.9	−1.2 (−2.1, −0.3)	0.387
IHTG content (%)[a]	19.5±4.1	9.6±3.0	−9.9 (−16.5, −3.3)	11.3±2.6	6.9±1.5	−4.5 (−7.9, −1.1)	0.540
Strength and cardiorespiratory fitness							
Sum of 1-RMs (kg)[b]	351±52	343±53	−8 (−38, 23)	267±23	301±26	35 (13, 56)	0.025
VO_{2peak} (L/min)[c]	3.0±0.4	2.9±0.4	−0.2 (−0.3, −0.1)	2.9±0.4	3.3±0.5	0.3 (0.0, 0.6)	0.007
Fasting blood chemistry							
Glucose (mg/dL)	98.0±3.2	91.9±2.2	−6.1 (−12.8, 0.6)	99.8±3.9	90.0±1.5	−9.8 (−17.0, −2.6)	0.113
Insulin (pmol/L)	186.7±29.6	116.1±21.6	−70.6 (−92.6, −48.5)	173.6±25.7	79.0±11.8	−94.5 (−133.2, −55.8)	0.009
HOMA-IR	7.7±1.4	4.3±0.8	−3.3 (−4.8, −1.8)	7.2±1.2	3.0±0.5	−4.2 (−6.1, −2.4)	0.001
HbA1c (%)	5.8±0.3	5.3±0.1	−0.5 (−1.0, 0.0)	5.7±0.2	5.2±0.1	−0.5 (−0.7, −0.3)	0.976
Triglyceride (mg/dL)	178±38	116±22	−62 (−116, −8)	140±12	103±10	−37 (−55, −18)	0.688
HDL cholesterol (mg/dL)	35.0±1.6	32.1±1.5	−2.9 (−5.7, −0.1)	38.5±2.7	37.3±3.0	−1.3 (−4.9, 2.4)	0.343
LDL cholesterol (mg/dL)	96±7	94±7	−2 (−20, 16)	116±11	108±9	−7 (−21, 6)	0.839
Total cholesterol (mg/dL)	167±11	150±10	−17.3 (−39.9, 5.4)	182±13	166±10	−15.6 (−30.8, −0.4)	0.450
NEFA (mmol/L)	0.48±0.06	0.44±0.04	−0.04 (−0.16, 0.08)	0.42±0.04	0.56±0.03	0.15 (−0.01, 0.31)	0.038
BCAAs (mmol/L)	764±41	634±47	−130 (−202, −58)	718±32	623±19	−94 (−161, −27)	0.630

Values before and after weight loss are expressed as mean±s.e.m. Change values are expressed as mean and 95% CIs within each group. Sum of 1-RMs denotes total weight lifted during leg press, knee flexion, chest press and seated row 1-RM strength tests. Two-sided ANCOVA without correction for multiple comparisons was used to determine between-group differences after adjusting for values before weight loss. [a]Diet+EX group *n*=7. [b]Diet-ONLY group *n*=6. [c]Diet-ONLY group *n*=7.

in the Diet-ONLY group (P = 0.014) (Fig. 4b and Extended Data Table 3). The Matsuda insulin sensitivity index (assessed using plasma glucose and insulin concentrations immediately before and for 2 h after glucose ingestion[27]) increased after weight loss in both groups, but the increase was ~4-fold greater in the Diet+EX group than in the Diet-ONLY group (P = 0.037) (Fig. 4c and Extended Data Table 3). The homeostasis model assessment of insulin resistance (HOMA-IR) values (assessed using fasting plasma glucose and insulin concentrations) decreased after weight loss in both the Diet-ONLY and Diet+EX groups, but the decrease was greater in the Diet+EX group than in the Diet-ONLY group (P = 0.001) (Table 1).

Skeletal muscle transcriptome

RNA sequencing (RNA-seq) was used to evaluate global changes in gene expression in the vastus lateralis muscle before and after weight loss. There were 877 differentially expressed genes (DEGs) (623 upregulated and 254 downregulated) in the Diet+EX group, and only one DEG (heme oxygenase 1, *HMOX1*), which was downregulated, in the Diet-ONLY group (Fig. 5a and Supplementary Table 1). The Database for Annotation, Visualization and Integrated Discovery (DAVID) was used to assess enrichment for Gene Ontology Biological Process (GO-BP) terms among the DEGs in the Diet+EX group. DEGs were highly enriched in 20 pathways (18 upregulated and 2 downregulated). Upregulated DEGs were predominantly enriched in many gene sets regulating mitochondrial energy metabolism and angiogenesis (Fig. 5b). Consistent with the pathway analysis, the expression of specific genes involved in mitochondrial biogenesis and energy metabolism (peroxisome proliferator-activated receptor γ coactivator 1α, *PPARGC1A*; mitochondrial fission factor, *MFF*; citrate synthase, *CS*; malate dehydrogenase 1, *MDH1*; cytochrome C somatic, *CYCS*; ATP synthase F1 subunit α, *ATP5F1A*; very long-chain acyl-CoA dehydrogenase, *ACADVL*; medium-chain acyl-CoA dehydrogenase, *ACADM*; lipoprotein lipase, *LPL*; and cluster of differentiation 36, *CD36*) and angiogenesis (vascular endothelial growth factor α, *VEGFA*; and vascular endothelial growth factor receptor 2, *KDR*) increased more in the Diet+EX group than in the Diet-ONLY group (all P < 0.05) (Fig. 5c,d and Extended Data Table 4). The downregulated DEGs in the Diet+EX group were enriched in the following two pathways: "Positive regulation of osteoblast differentiation" and "Skeletal muscle contraction."

Plasma proteome

A total of 4,684 plasma proteins were identified before and after weight loss by using the aptamer-based SomaScan proteomics assay (SomaLogic) to assess whether (1) the circulating proteome was altered by weight loss with and without concomitant exercise, and (2) alterations in the skeletal muscle gene transcriptome were associated with a corresponding change in the encoded circulating proteins. There were 111 (50 upregulated and 61 downregulated) and 164 (77 upregulated and 87 downregulated) proteins with differential abundance after weight loss in the Diet-ONLY and Diet+EX groups, respectively (Fig. 6a). Using the protein enrichment analysis of GO-BP, we found that the most enriched pathways were upregulated in the Diet+EX group but were downregulated in the Diet-ONLY group (Fig. 6b). GO-BP terms related to autophagy and ribose and purine biosynthesis were the most significantly upregulated pathways in the Diet+EX group, whereas these pathways were downregulated

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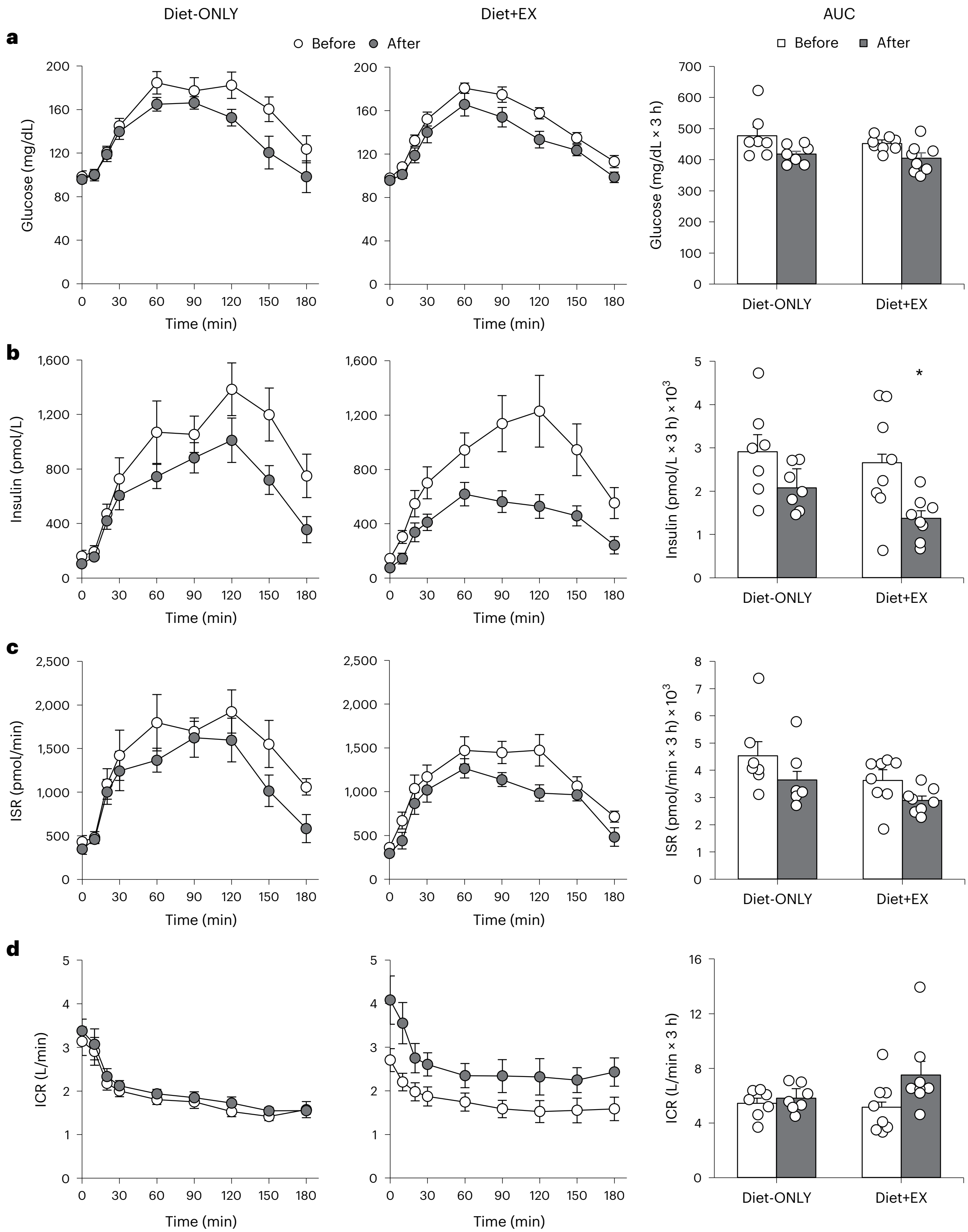

Fig. 1 | Glucose and insulin responses to glucose ingestion. Plasma concentrations and AUCs for glucose (**a**), insulin (**b**), ISR (**c**) and ICR (**d**) in response to glucose ingestion in the Diet-ONLY ($n = 7$) and Diet+EX ($n = 8$) groups before (white circles) and after (gray circles) ~10% weight loss. Data are mean ± s.e.m. ANCOVAs with baseline values as covariates were used to compare weight loss-induced changes in AUCs between groups. Two-sided ANCOVA without correction for multiple comparisons: *$P = 0.008$ versus Diet-ONLY value.

in the Diet-ONLY group (Fig. 6b). We then analyzed Gene Ontology Cellular Component (GO-CC) terms to better understand the source of differentially abundant proteins in plasma. The most significantly enriched cellular component in the Diet+EX group was an upregulation of "Mitochondrion," which was markedly downregulated in the Diet-ONLY group. The cellular component "Mitochondrial matrix" was also downregulated in the Diet-ONLY group and upregulated in the Diet+EX group. In addition, 12 circulating proteins involved in mitochondrial biology and energy metabolism and the genes in muscle encoding those proteins were increased in the Diet+EX group (Fig. 6c) but not in the Diet-ONLY group. These data suggest that the changes in skeletal muscle gene expression induced by Diet+EX produce several proteins that are released into the circulation and increase their plasma concentrations.

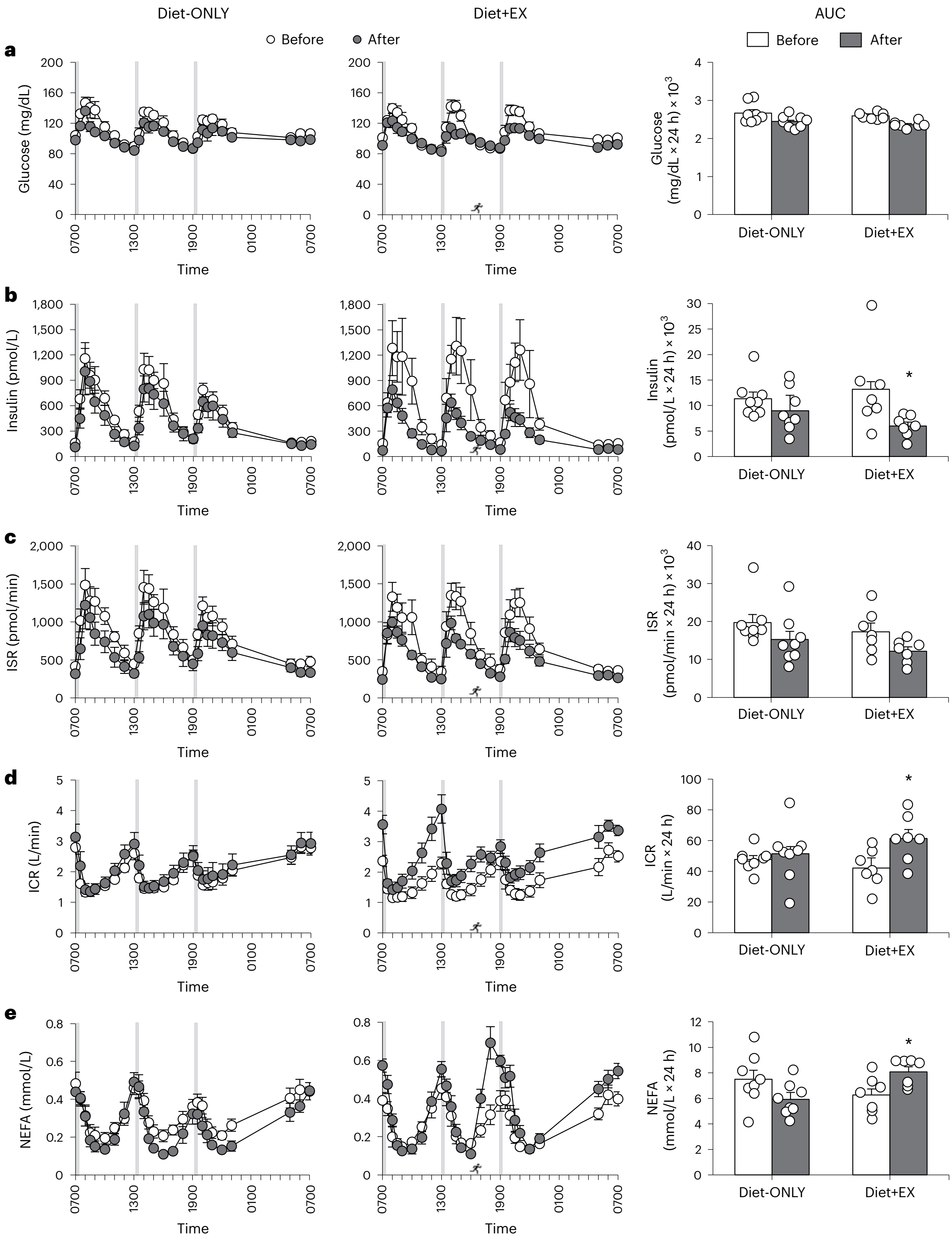

Fig. 2 | 24-h plasma glucose, insulin and NEFA dynamics. Plasma concentrations and AUCs for glucose (**a**), insulin (**b**), ISR (**c**), ICR (**d**) and NEFA (**e**) obtained over 24 h in the Diet-ONLY ($n = 8$) and Diet+EX ($n = 7$) groups before (white circles) and after (gray circles) ~10% weight loss. Gray vertical bars and running figure indicate time of mixed-meal ingestion and a 60-min bout of exercise performed in the Diet+EX group after ~10% weight loss, respectively. A detailed timeline is presented in Extended Data Fig. 1. Data are mean ± s.e.m. ANCOVAs with baseline values as covariates were used to compare weight loss-induced changes in AUCs between groups. Two-sided ANCOVA without correction for multiple comparisons: *$P = 0.030$, 0.009 and 0.003 versus Diet-ONLY value for plasma insulin, ICR and plasma NEFA AUCs, respectively.

Gut microbiome

To assess the effect of the two interventions on the composition of the gut microbiome, we used 16S rRNA amplicon sequencing to assess the microbiota in fecal samples collected before (T0), during (T1 and T2, which were obtained at about 1 month and 2 months after starting the intervention when participants had lost 5.2 ± 0.7% and 8.6 ± 0.8% body weight, respectively), and while weight-stable at the end of the intervention (T3, which were obtained ~4–5 months

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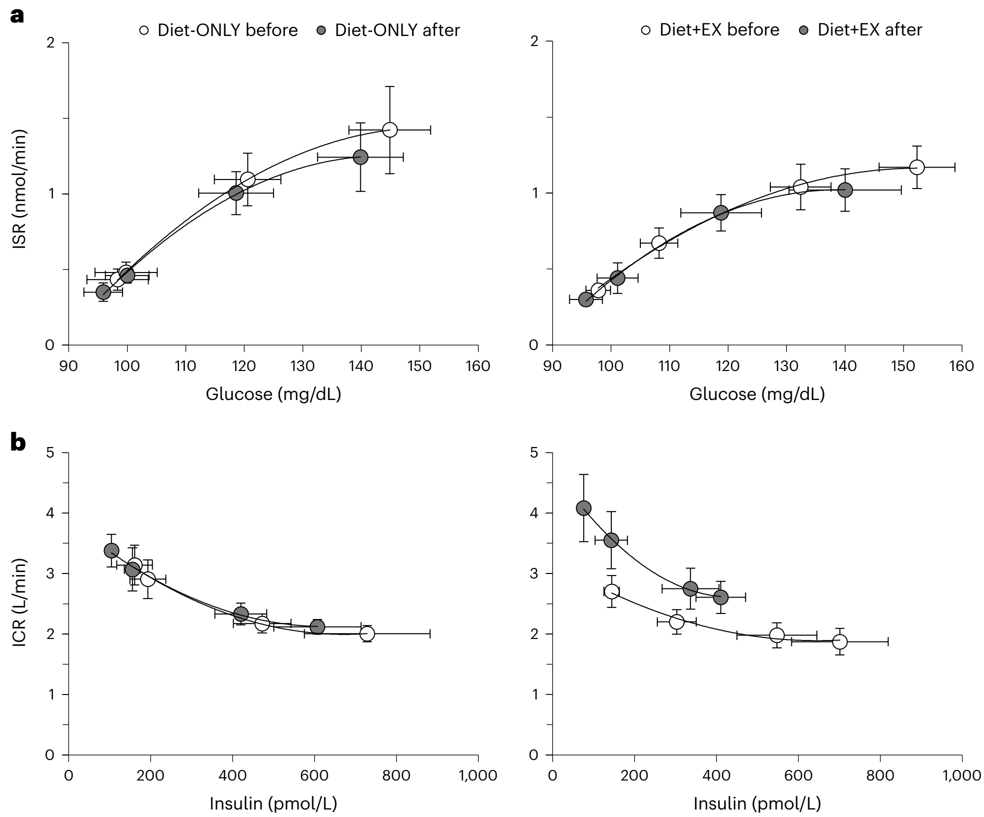

Fig. 3 | β-cell function and insulin clearance during the first 30 min of the OGTT. Relationships between plasma glucose concentration and ISR (**a**), and plasma insulin concentration and ICR (**b**) before and during the first 30 min after ingesting 75 g of glucose in the Diet-ONLY (*n* = 7) and Diet+EX (*n* = 8) groups before (white circles) and after (gray circles) ~10% weight loss. Data are mean ± s.e.m.

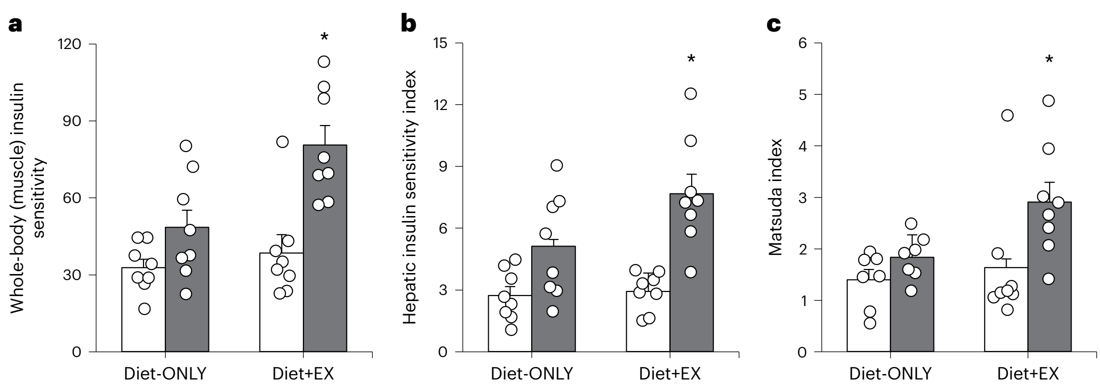

Fig. 4 | Effect of Diet-ONLY and Diet+EX on insulin sensitivity. Whole-body (primarily muscle) insulin sensitivity (glucose disposal rate per kilogram FFM divided by plasma insulin concentration during the hyperinsulinemic-euglycemic clamp procedure) (**a**), hepatic insulin sensitivity index (the reciprocal of the product of endogenous glucose production rate per kilogram FFM and plasma insulin concentration during basal conditions) (**b**) and the Matsuda insulin sensitivity index (10,000 divided by the square root of the product of plasma glucose and insulin concentrations during the basal state and mean plasma glucose and insulin concentrations at 30, 60, 90 and 120 min during the OGTT) (**c**) in the Diet-ONLY (*n* = 8 in **a** and **b**, and *n* = 7 in **c**) and Diet+EX (*n* = 8 in **a**, **b** and **c**) groups before (white bars) and after (gray bars) ~10% weight loss. Data are mean ± s.e.m. ANCOVAs with baseline values as covariates were used to compare weight loss-induced changes in AUCs between groups. Two-sided ANCOVA without correction for multiple comparisons: **P* = 0.006, 0.014 and 0.037 versus Diet-ONLY value for whole-body insulin sensitivity, hepatic insulin sensitivity and Matsuda insulin sensitivity index, respectively.

after starting the intervention). Alpha diversity was unchanged, and beta diversity changed in the same direction and to a similar magnitude across T0, T1, T2 and T3 in both the Diet-ONLY and Diet+EX groups (Extended Data Fig. 2), so data were analyzed separately and with both treatment groups combined to enhance the ability to identify diet-induced and weight loss-induced changes in the gut microbiome.

Alpha diversity, measured as either Chao richness or Shannon entropy of the amplicon sequence variants (ASVs), was not different at any time point during the intervention than the values at baseline in the combined analysis (Fig. 7a), or in either group (Extended Data Fig. 2a). There was an overall effect of time on beta diversity, assessed by distance-based redundancy analysis (ordination plot) and Adonis through permutational multivariate analysis of variance (PERMANOVA;

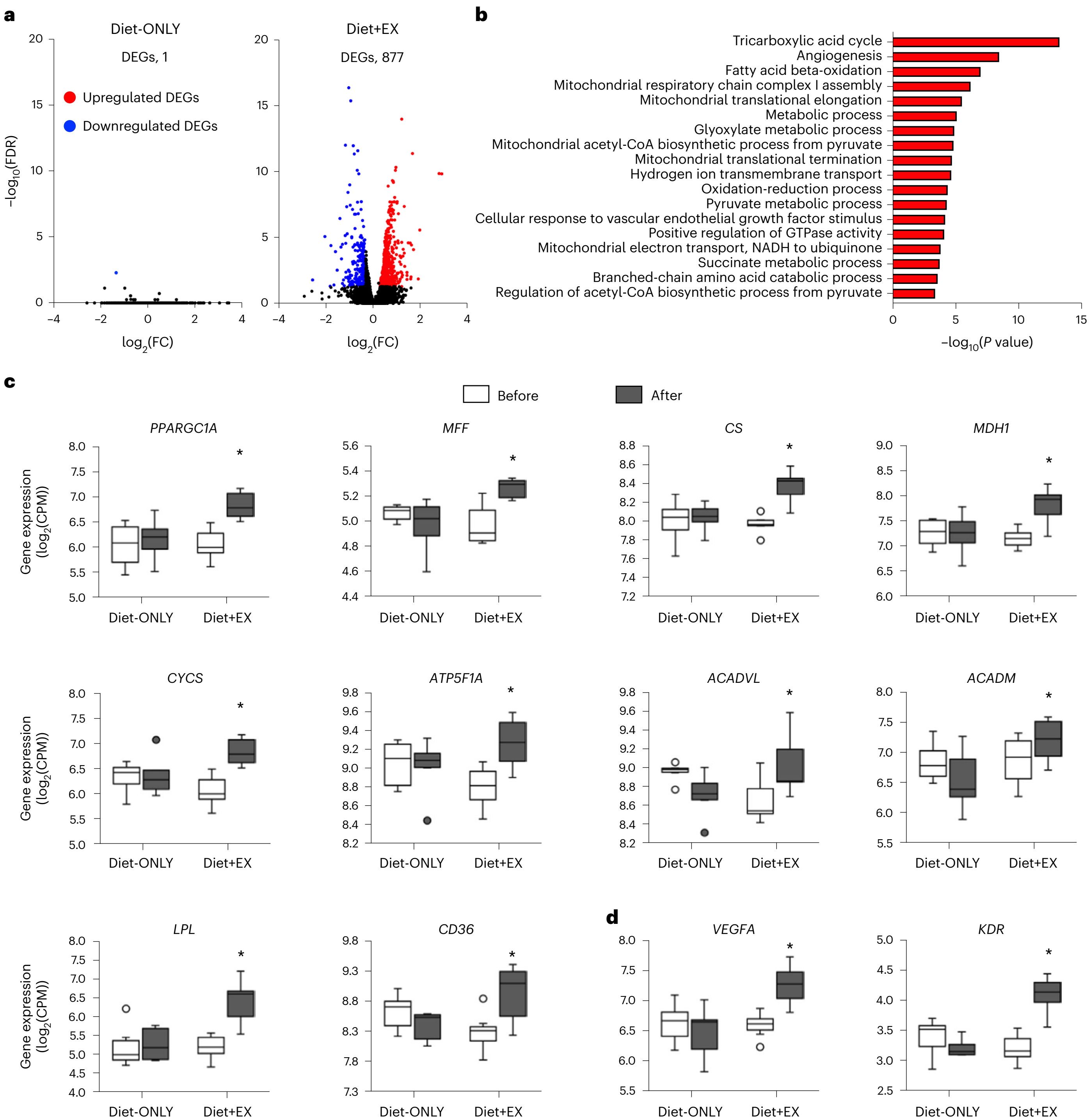

Fig. 5 | Skeletal muscle RNA-seq. Skeletal muscle tissue obtained in the Diet-ONLY ($n = 6$) and Diet+EX ($n = 7$) groups before and after weight loss were evaluated using RNA-seq. **a**, Volcano plots of skeletal muscle RNA-seq data with $\log_2(\mathrm{FC})$ (x axis) and two-sided $-\log_{10}(\mathrm{FDR})$ (y axis). The number of DEGs ($|\log_2(\mathrm{FC})| > 0.3$ and two-sided FDR < 0.05) between before and after weight loss in each group are shown. **b**, Biological pathways significantly (two-sided FDR < 0.01) enriched with upregulated DEGs between before and after weight loss in the Diet+EX group. **c**,**d**, Expression of key genes that regulate mitochondrial biogenesis and energy metabolism (**c**), and expression of genes involved in regulating angiogenesis (**d**) in the Diet-ONLY and Diet+EX groups before (white bars) and after (gray bars) ~10% weight loss. Data in **c** and **d** are median values (central horizontal line), 25th and 75th percentiles (box), minimum and maximum values (vertical lines) and outliers (circles), and were analyzed using ANCOVAs with baseline values as covariates to compare weight loss-induced changes between groups. Two-sided ANCOVA without adjustment for multiple comparisons: *$P = 0.001, 0.009, <0.001, 0.007, 0.003, 0.030, 0.024, 0.001, 0.006, 0.001, 0.004$ and <0.001 versus Diet-ONLY value for *PPARGC1A*, *MFF*, *CS*, *MDH1*, *CYCS*, *ATP5F1A*, *ACADVL*, *ACADVM*, *LPL*, *CD36*, *VEGFA* and *KDR*, respectively.

$P < 0.05$) of weighted UniFrac distances (Fig. 7b and Extended Data Fig. 2b). Gut microbial composition changed from T0 to T1 of the intervention (PERMANOVA, $P < 0.05$), without further changes at T2 and T3 despite continued weight loss (PERMANOVA, $P > 0.05$ for each pairwise comparison among T1, T2 and T3), demonstrating that most of the change in the composition of the gut microbiome occurred within the first month after beginning the intervention in both the combined and by-group analyses (Fig. 7b and Extended Data Fig. 2b). In addition,

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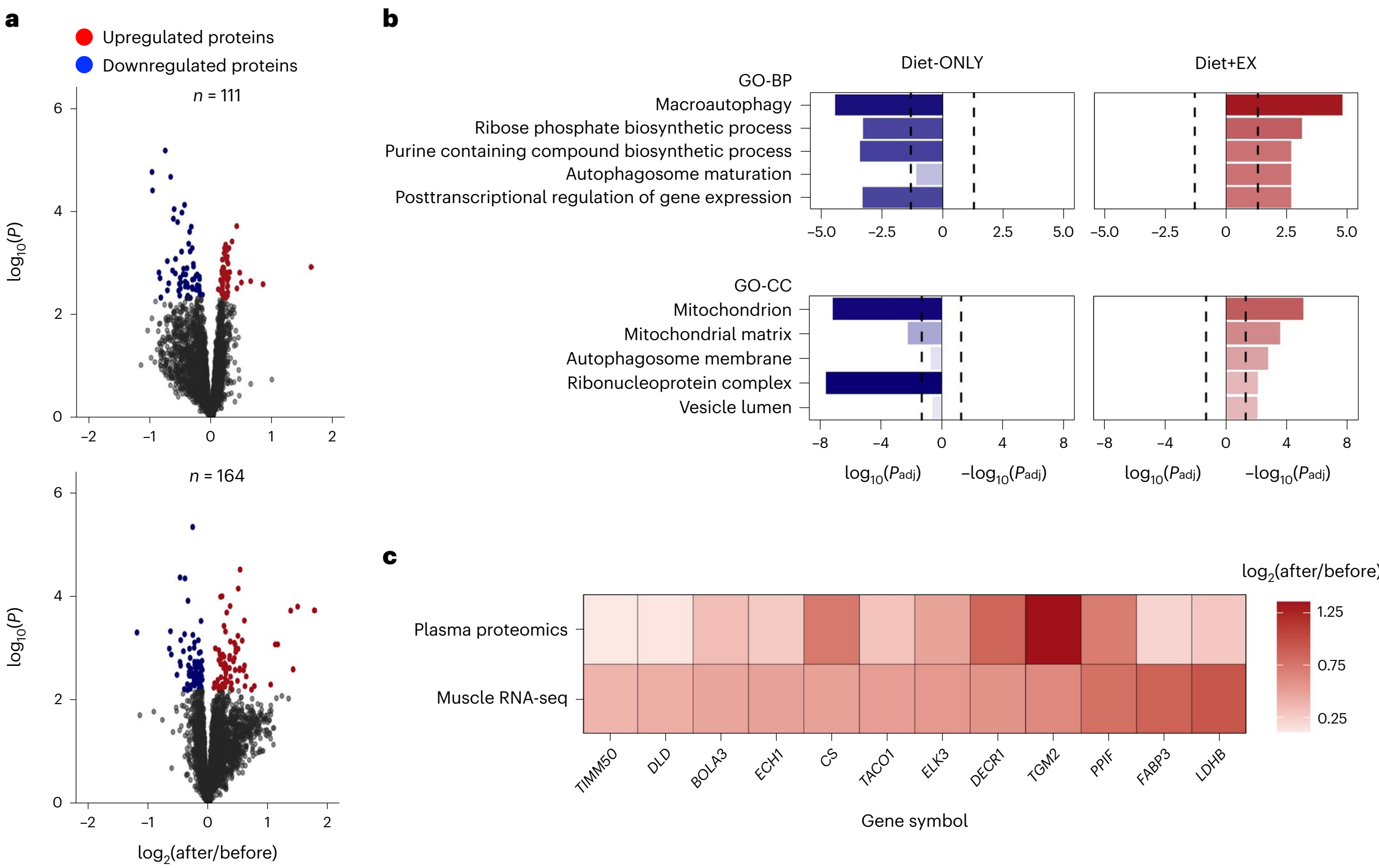

Fig. 6 | Changes in the plasma proteome. a, Volcano plots of proteins with differential abundance after weight loss in the Diet-ONLY ($n = 8$) and Diet+EX groups ($n = 8$). **b**, Functional enrichment analyses were performed to identify biological pathways significantly enriched (two-sided $P_{adj} < 0.20$) with proteins with differential abundance after weight loss. The top five most significantly enriched pathways in the GO-BP and GO-CC databases in the Diet+EX group and the corresponding changes in the Diet-ONLY group are shown. Vertical dashed lines represent two-sided $\log_{10}(P_{adj}) < 0.05$. **c**, Plasma proteins related to mitochondrial biology and energy metabolism (mitochondrial import inner membrane translocase subunit TIM50, *TIMM50*; mitochondrial dihydrolipoyl dehydrogenase, *DLD*; bolA-like protein 3, *BOLA3*; enoyl-CoA hydratase 1, *ECH1*; citrate synthase, *CS*; translational activator of cytochrome c oxidase 1, *TACO1*; ETS domain-containing protein Elk-3, *ELK3*; 2,4-dienoyl-CoA reductase 1, *DECR1*; transglutaminase 2, *TGM2*; mitochondrial peptidyl-prolyl cis-trans isomerase F, *PPIF*; fatty acid binding protein 3, *FABP3*; lactate dehydrogenase B, *LDHB*) that significantly increased ($P_{adj} < 0.20$) after the intervention in conjunction with a significant (two-sided FDR $P < 0.05$) increase in skeletal muscle expression of the encoding genes in the Diet+EX group. Differential abundance of each protein from before to after weight loss in each group was tested using empirical Bayes moderated paired *t*-tests, *P* values were adjusted using the Benjamini–Hochberg method, and a two-sided $P_{adj} < 0.20$ was considered significant.

community changes over time were not different between Diet+EX and Diet-ONLY (group × time point interaction using PERMANOVA, $P = 0.458$; Extended Data Fig. 2b).

Changes in ASVs were analyzed using linear mixed-effects negative binomial model analysis comparing ASVs at T1 and T0, which coincided with the largest difference in beta diversity between time points (Supplementary Tables 2 and 3). The ten ASVs with the largest increase and the ten ASVs with the largest decrease from T0 to T1 were identified (Fig. 7c). The log-ratio[28] of the ASVs that changed from T0 to T1 remained greater than those of T0 at T2 and T3 in the pooled analysis (Fig. 7d), with no difference in the log-ratio of these ASVs between the Diet+EX and Diet-ONLY groups ($P = 0.603$ by linear mixed-effects model) (Extended Data Fig. 2c).

Adverse events

There were very few study-related adverse events in either group. One participant in the Diet-ONLY group experienced anxiety during baseline magnetic resonance imaging (MRI) scanning. In the Diet+EX group, one participant experienced anxiety during baseline MRI scanning, one had a syncopal episode after the muscle biopsy procedure, one reported a headache of unknown cause and one reported significant muscle soreness after a bout of exercise the previous day.

Discussion

The purpose of the present study was to assess the potential clinical and metabolic importance of incorporating multimodal exercise training as part of a weight loss therapy program in people with obesity and prediabetes. Accordingly, we evaluated the effect of 10% diet-induced weight loss with and without concomitant multimodal exercise training on (1) whole-body (muscle) insulin sensitivity assessed using the hyperinsulinemic-euglycemic clamp procedure (primary outcome); (2) cardiometabolic secondary outcomes (hepatic insulin sensitivity, body composition, cardiorespiratory fitness, muscle strength, metabolic responses to glucose ingestion and repeated mixed-meal consumption, β-cell function and insulin kinetics); and (3) putative mechanisms responsible for weight loss-induced and exercise-induced improvements in insulin action, namely skeletal muscle transcriptomics, plasma proteomics, circulating markers of inflammation, plasma BCAA and NEFA concentrations, and the gut microbiome. Our data demonstrate that 10% weight loss induced by diet alone or diet plus exercise training improved multiorgan insulin sensitivity and decreased basal and postprandial insulin concentrations. However, we found that the increase in whole-body insulin sensitivity was much greater after weight loss induced by diet and exercise than matched weight loss

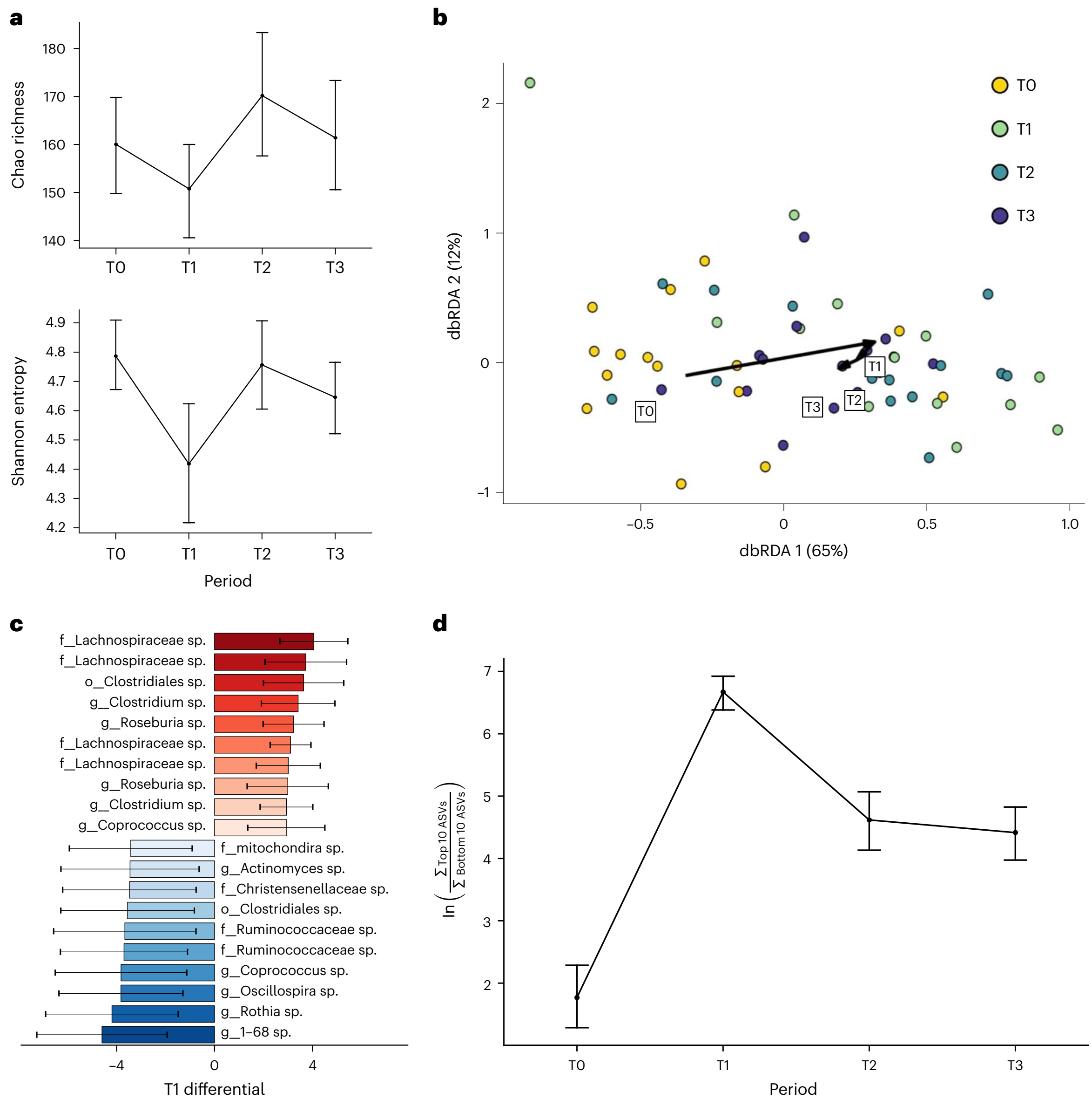

Fig. 7 | Changes in gut microbiota. a, Measures of alpha diversity using both Chao richness (top) and Shannon entropy (bottom), before (T0) and at ~1 month (T1, 5.2 ± 0.7% weight loss), ~2 months (T2, 8.6 ± 0.8% weight loss) and ~4–5 months (T3, 10.5 ± 2.7% weight loss) after starting the intervention in all participants (all panels are $n = 15$; Diet-ONLY $n = 7$ and Diet+EX $n = 8$). Data are mean ± s.e.m. **b**, Beta diversity using distance-based redundancy analysis (dbRDA) of the weight UniFrac distance, conditional on the between-person variability. **c**, Differential abundance results of the ten ASVs that increased the most and the ten ASVs that decreased the most from T0 to T1 with treatment groups combined. Red and blue bars represent posterior mean value ± s.d. **d**, Log-ratio analysis of top ten ASVs and bottom ten ASVs shown in **c** over time with both groups combined. Data are log-ratio mean ± s.e.m.

induced by diet alone. We also found that changes in our secondary outcomes, namely, the increase in hepatic insulin sensitivity and the decreases in basal, postprandial and 24-h plasma insulin concentrations, were greater after weight loss induced by diet and exercise than matched weight loss induced by diet alone. However, the conclusions from these secondary outcomes should be considered "hypothesis generating," and the statistical analysis of differences between groups was not corrected for multiple testing. An unbiased assessment of skeletal muscle transcriptomics and the plasma proteome suggests that weight loss induced by diet plus exercise caused marked remodeling of the mitochondrial and vascular network of skeletal muscle by increasing mitochondrial biogenesis, mitochondrial content and angiogenesis. Together, these data demonstrate that multimodal exercise has profound additional beneficial effects on multiorgan insulin sensitivity and insulin kinetics, beyond weight loss alone, in people with obesity and prediabetes. These results underscore the importance of including exercise training as a component of obesity management, in addition to a calorie-deficit diet, even though exercise per se typically has minimal effects on body weight.

The timing of testing in relation to the last bout of exercise has an important influence on the assessment of insulin sensitivity. Even a single 45-min session of endurance exercise improves insulin sensitivity measured 48 h later, and additional short-term (6 weeks) exercise training in the same participants caused a further increase in insulin sensitivity[3]. However, the increase in insulin sensitivity after an acute bout of exercise or exercise training decreases over time and is usually back to baseline within 72–96 h[29,30]. Therefore, evaluating metabolic function several days after the last bout of exercise could underestimate the beneficial effect of an exercise training program on insulin action. In the present study, we used the hyperinsulinemic-euglycemic clamp procedure, in conjunction with stable isotopically labeled glucose tracer infusion, to provide a robust assessment of hepatic and whole-body (muscle) insulin sensitivity at ~17.5 h and ~21 h after the last bout of exercise, respectively. Our data demonstrate that regular exercise training has considerable additive effects in improving insulin sensitivity compared with weight loss alone, likely owing to both the chronic and acute effects of exercise.

The mechanism responsible for the additive effect of exercise on insulin-stimulated glucose uptake is likely related to exercise-induced changes in skeletal muscle biology. We found that skeletal muscle expression of genes involved in mitochondrial energy metabolism and substrate oxidation increased more in the Diet+EX group than in the Diet-ONLY group, which is consistent with results from previous studies[31,32]. Moreover, an analysis of the plasma proteome reinforced the skeletal muscle transcriptomic results by demonstrating an increase in several circulating proteins involved in skeletal muscle energy metabolism and mitochondrial biology after weight loss in the Diet+EX group compared with the Diet-ONLY group. It has been proposed that decreased skeletal muscle mitochondrial content and function are involved in the pathogenesis of muscle insulin resistance[33,34], which suggests that the increase in muscle mitochondrial content or function in the Diet+EX group could have contributed to the greater increase in muscle insulin sensitivity compared with the Diet-ONLY group. It is also likely that the potent effects of exercise on insulin sensitivity were mediated by additional factors that increase skeletal muscle glucose delivery, uptake and metabolism, such as exercise-induced increases in muscle capillary density and capillary:fiber ratio[35], tissue perfusion[36], the number of plasma membrane GLUT4 glucose transporters[37], and insulin-stimulated muscle glycogen synthesis[3].

We evaluated the effect of matched weight loss with and without exercise training on specific circulating factors that are purportedly involved in the pathogenesis of insulin resistance, namely, plasma BCAAs, cytokines and NEFAs[38]. Weight loss decreased fasting plasma BCAA concentrations in both the Diet+EX and Diet-ONLY groups, without a difference in the magnitude of the reduction in plasma BCAAs between groups. Fasting plasma cytokine concentrations were not affected by weight loss in the Diet-ONLY and Diet+EX groups apart from plasma PAI-1, which decreased by ~55% in both groups after weight loss. We recently found that plasma PAI-1 concentrations are inversely associated with hepatic and skeletal muscle insulin sensitivity[39], supporting the potential importance of PAI-1 in regulating insulin sensitivity. Plasma 24-h NEFA concentration AUC decreased after weight loss in the Diet-ONLY group, but increased after weight loss in the Diet+EX group, presumably because of exercise-induced stimulation of adipose tissue lipolytic activity[40]. These findings suggest that reductions in plasma BCAA and PAI-1 concentrations contribute to weight loss-induced improvements in insulin sensitivity, but that plasma BCAAs, cytokines and NEFAs do not explain the greater improvement in insulin sensitivity observed after moderate weight loss with exercise training compared with moderate weight loss alone.

Weight loss with or without exercise training did not affect β-cell function, assessed as the ISR in relation to plasma glucose concentration during the first 30 min after glucose ingestion. Nonetheless, ISRs after glucose ingestion or during 24 h with mixed-meal consumption were lower after weight loss in both groups, owing to a decreased stimulus for insulin secretion caused by lower postprandial plasma glucose concentrations. Although basal, postprandial and 24-h plasma insulin concentrations decreased after weight loss in both groups, the decrease was greater in the Diet+EX group than in the Diet-ONLY group. The difference in plasma insulin between groups was related to an increase in ICR after weight loss in the Diet+EX group, which did not change after weight loss in the Diet-ONLY group. The mechanism responsible for the increase in ICR in the Diet+EX group is likely related to exercise training-induced increases in insulin extraction by the liver[41,42] and skeletal muscle[43]. Together, these data demonstrate that weight loss with or without exercise does not affect β-cell function in people with obesity and prediabetes, and that diet-induced weight loss decreases plasma insulin concentrations by decreasing ISR, whereas diet-induced weight loss plus exercise training decreases plasma insulin concentrations by decreasing ISR and increasing ICR.

Distinct alterations in the composition of the gut microbiome are associated with obesity, metabolic syndrome and type 2 diabetes[44]. Moreover, transplantation of fecal microbiota from lean healthy donors to people with obesity and metabolic syndrome improves insulin sensitivity in the recipients, suggesting that the composition of the gut microbiome can directly affect metabolic function[45,46]. Therefore, we evaluated fecal samples obtained at baseline, ~1 month and ~2 months during the weight loss intervention, and at ~4–5 months, after participants lost about 10% of their body weight and were weight-stable for several weeks. Our data demonstrate that consumption of a plant-forward diet causes marked changes in the composition of the gut microbiome beta diversity within 1 month of initiating diet therapy, without further changes thereafter, despite continued weight loss. Moreover, the change in microbial diversity was not different between the Diet-ONLY and Diet+EX groups. These results demonstrate the profound effect of dietary composition on beta diversity that is not further modified by regular exercise. The changes in the global microbial communities were reflected by specific microbial taxa that responded to the dietary intervention after ~1 month, a change that persisted with continued intervention in both groups. However, we are unable to determine whether these alterations in the gut microbiome contributed to the improvements in insulin sensitivity observed in our participants or were simply associated with the change in dietary composition without influencing insulin action.

Our study has some limitations. First, the participants in the Diet+EX group completed all postintervention metabolic studies the day after completing an exercise bout, so we cannot determine the separate contributions from chronic and recent exercise to the greater increase in insulin sensitivity observed in the Diet+EX group compared with the Diet-ONLY group. Nonetheless, our study design represents an assessment of the effect of regular exercise and avoids the confounding effect of detraining on insulin sensitivity. Second, our study was conducted in people with obesity and prediabetes who participated in a short-term (~5 months) supervised multimodal exercise training program and who were given a plant-forward diet because this type of diet enhances weight loss and has beneficial effects on cardiometabolic outcomes[22–24]. Therefore, we do not know if similar results would be obtained in other patient populations, or after treatment with different types of dietary and exercise training regimens. Third, the intensity of the lifestyle intervention in our study (all meals provided and supervised exercise training), which was needed to reliably test our hypothesis, is not feasible for most people in the "real world." Fourth, our study was not a randomized trial. However, to help ensure balance between the Diet-ONLY and Diet+EX groups, the groups were studied simultaneously to avoid the limitations of historical controls, and the eligibility and exclusion criteria were identical for both the Diet-ONLY and Diet+EX groups. Accordingly, the key baseline characteristics were similar in the Diet-ONLY and Diet+EX groups. Fifth, our study included many secondary outcomes that should be considered "hypothesis generating," and the statistical analysis of differences between groups was not corrected for multiple testing. However, the secondary outcomes that improved more in the Diet+EX group than in the Diet-ONLY group (such as cardiorespiratory fitness, muscle strength, and muscle expression of genes related to mitochondria and energy metabolism) are consistent with the expected effects of exercise training, despite the increased chance of a type 1 statistical error.

In summary, the results from the present study demonstrate that multimodal exercise, conducted in conjunction with diet-induced weight loss in people with obesity and prediabetes, caused greater improvement in whole-body (muscle) insulin sensitivity and in key secondary outcomes, including hepatic insulin sensitivity, cardiorespiratory fitness and muscle strength, than matched weight loss achieved solely by calorie restriction. These findings underscore the importance of targeting skeletal muscle biology in regulating metabolic health and physical function. Accordingly, exercise training should be incorporated as part of any weight loss program to maximize the cardiometabolic benefits of lifestyle therapy.

Methods

Participants

The overall study protocol is shown in Extended Data Fig. 1. Twenty-one adults with obesity and prediabetes participated in one of two interventions: (1) 10% weight loss induced by calorie restriction alone (Diet-ONLY; $n = 11$, 8 self-reported as females at birth), or (2) 10% weight loss induced by calorie restriction plus exercise training (Diet+EX; $n = 10$, 7 self-reported as females at birth). Participants provided written, informed consent before participating in this study, which was approved by the Institutional Review Board within the Human Research Protection Office at Washington University School of Medicine in St. Louis, MO, and registered at ClinicalTrials.gov (NCT02706262 and NCT02706288). All participants were reimbursed for time and effort. Participants in the Diet-ONLY group were also part of a larger ongoing clinical trial comparing the metabolic effects of 10% weight loss induced by three different dietary interventions without exercise training (NCT02706262). The participants in each group were enrolled in their respective trials concurrently between April 2016 and November 2018. The Diet-ONLY and Diet+EX groups were studied simultaneously to avoid the limitations of historical controls, and the eligibility and exclusion criteria were identical for both groups. All participants were recruited using the Volunteers for Health database at Washington University School of Medicine and by local postings between April 2016 and November 2018. Study visits were conducted in the Clinical Translational Research Unit (CTRU) and the Center for Clinical Imaging Research at Washington University School of Medicine. All participants completed a comprehensive screening evaluation that included a medical history and physical examination, standard blood tests, HbA1c, and an oral glucose tolerance test (OGTT) to determine eligibility. The following inclusion criteria were required: body mass index (BMI) of 30.0–49.9 kg/m^2, and prediabetes (HbA1c ≥ 5.7–6.4% or fasting plasma glucose concentration of 100–125 mg/dL, or 2-h OGTT plasma glucose concentration of 140–199 mg/dL). In addition, all participants had evidence of whole-body insulin resistance, defined as a glucose infusion rate of <8 mg/kg FFM/min during a hyperinsulinemic-euglycemic clamp procedure[47]. Potential participants who had a history of diabetes, liver disease other than NAFLD, were taking medications that could affect the study outcomes, consumed excessive amounts of alcohol (>21 units of alcohol per week for men and >14 units of alcohol per week for women), or were pregnant or lactating were excluded. A total of 16 participants completed baseline and post-weight loss testing procedures and were included in the comparison of the Diet-ONLY ($n = 8$, 4 females; mean age, 43 ± 3 years) and Diet+EX ($n = 8$, 6 females; mean age, 46 ± 3 years) groups (Extended Data Fig. 3). Data collection was not performed blind to the conditions of the experiments.

Body composition analyses

Body fat mass, FFM and appendicular lean mass were determined using dual-energy X-ray absorptiometry (Lunar iDXA, GE Healthcare). IHTG content was determined using MRI (3 T superconducting magnet, Siemens)[48]. One participant in the Diet+EX group was unable to complete the MRI scans due to claustrophobia.

Cardiorespiratory fitness and maximal strength assessment

Cardiorespiratory fitness (VO_{2peak}) was determined using indirect calorimetry (ParvoMedics) during a progressive treadmill test, in which speed and grade were adjusted in 2-min stages until volitional exhaustion. Maximal effort during the test was defined as achieving at least two of the following criteria: (1) a respiratory exchange ratio at maximal exercise of >1.15, (2) a leveling off of heart rate with no more than a 3-beat/min increase between the final two workloads, and (3) a plateau in oxygen consumption with <0.2 L/min increase between the final two workloads. On a separate occasion, participants attended an orientation session to become familiar with the weight-lifting equipment (multistation weight machine, Hoist Fitness Systems) and to receive instruction on the correct techniques for the following exercises (all bilateral): leg press, knee flexion, chest press and seated row. Participants returned to the research unit ~1 week later to determine maximal strength (1-RM) for each of these exercises[49]. For technical reasons, data were not obtained for cardiorespiratory fitness and maximal strength from one and two participants, respectively, in the Diet-ONLY group.

3-h oral glucose tolerance test

Participants were admitted to the outpatient unit of the CTRU at Washington University School of Medicine at 0700 hours after participants fasted for approximately 11 h overnight at home. An intravenous catheter was inserted into an antecubital or hand vein for serial blood sampling. Plasma glucose, insulin and C-peptide concentrations were determined 15, 10 and 5 min before and 10, 20, 30, 60, 90, 120, 150 and 180 min after consuming a 75-g glucose beverage. The average of the three baseline samples obtained at 15, 10 and 5 min before consuming the beverage was used as the 0-min value for glucose, insulin and plasma C-peptide concentrations. Due to technical issues, one participant in the Diet-ONLY group did not complete the OGTT after weight loss.

24-h blood sampling and assessment of insulin sensitivity

The timeline for the inpatient studies is shown in Extended Data Fig. 1d. Participants were admitted to the CTRU at 1800 hours for ~48 h and consumed a standard meal containing one-third of their estimated energy requirements[25] at 1900 hours. At 0630 hours on the morning of day 2, a catheter was inserted into an antecubital vein for 24-h serial blood sampling. Blood samples were obtained hourly from 0700 hours to 2300 hours on day 2 and from 0500 hours to 0700 hours on day 3; additional blood samples were obtained 30 min and 90 min after each meal, which were given at 0700 hours, 1300 hours and 1900 hours. Each meal contained one-third of the participant's estimated energy requirement[25]. After the evening meal was consumed on day 2, participants fasted until the end of the hyperinsulinemic-euglycemic clamp procedure conducted on day 3. At 0700 hours on day 3, a primed (8.0 μmol/kg) continuous (0.08 μmol/kg/min) intravenous infusion of [U-^{13}C]glucose (Cambridge Isotope Laboratories) was started and maintained for 210 min (basal period). An additional catheter was inserted into a radial artery to obtain arterial blood samples. At the end of the basal period, the infusion of [U-^{13}C]glucose was stopped, and an insulin infusion was initiated at a rate of 50 mU/m^2/min (initiated with a two-step priming dose of 200 mU/m^2 BSA/min for 5 min followed by 100 mU/m^2/min for 5 min) for an additional 210 min. Euglycemia (~100 mg/dL) was maintained by variable rate infusion of 20% dextrose enriched to ~1% with [U-^{13}C]glucose. Blood samples were obtained before beginning the tracer infusion and every 6–7 min during the final 20 min of the basal and insulin infusion periods. Due to technical issues, one participant in the Diet+EX group did not complete the 24-h blood sampling after weight loss.

Skeletal muscle biopsy

Skeletal muscle tissue was obtained during the basal period of the clamp procedure before and after the intervention in the Diet-ONLY ($n = 6$) and Diet+EX ($n = 8$) groups. Muscle tissue was obtained from the quadriceps femoris using local anesthesia (lidocaine 2%) and Tilley

Henckel forceps. Tissue samples were immediately rinsed in ice-cold saline and frozen in liquid nitrogen before being stored at −80 °C until processing for RNA-seq.

Fecal collections

Fecal samples were collected at home, frozen at −20 °C and then transported to the CTRU in a polystyrene foam cooler containing frozen cold packs within 36 h of production. Samples were then stored at −80 °C until processing for fecal 16S rRNA amplicon sequencing. During baseline (T0) and postintervention (T3) testing, three fecal samples were collected ~7 days apart, and the median of these replicates is reported. The T3 samples were collected when participants were weight-stable for 1–4 weeks (Extended Data Fig. 2). Additional fecal samples were collected during progressive weight loss after approximately 1 month (T1) and 2 months (T2) on the calorie-restricted diet. Baseline samples at T0 were collected while participants were on their habitual diet, and samples collected at T1, T2 and T3 were collected while the participants were on the intervention diet.

Diet and exercise interventions

After completing baseline testing, individuals participated in a supervised weight loss program involving weekly individual dietary and behavioral education sessions with all food provided as packed-out meals. All participants consumed a plant-forward, "Pritikin-type" diet (low in fat, sodium and refined carbohydrates, and high in complex carbohydrates, primarily derived from plant-based sources (vegetables, fruit, whole grains, seeds and legumes)) that provided ~70% of energy as carbohydrates, ~15% as fats and ~15% as protein. This specific diet was selected because it has demonstrated considerable beneficial effects on glycemic control and other cardiometabolic abnormalities[22,24]. The initial daily energy content of the diet provided 75% of estimated energy requirements[25]; subsequent energy intake was adjusted weekly as needed to achieve a 0.5–1% weight loss per week until ~10% weight loss was achieved.

Participants in the Diet+EX group also performed six 1-h exercise training sessions per week throughout the weight loss intervention; four sessions per week were conducted under the supervision of an exercise physiologist, and two sessions were conducted at home. Exercise sessions were performed at the time of day most convenient for each participant, which was usually in the afternoon or early evening. The prescribed exercise training was designed in accordance with the American College of Sports Medicine guidelines on physical activity and weight loss[50]. Supervised exercise sessions consisted of two sessions of aerobic exercise, one session of high-intensity interval training (HIIT) and one session of resistance training per week. During aerobic exercise sessions, participants were free to choose between exercising on treadmills, stationary cycle ergometers or elliptical trainers at 65–75% of heart rate reserve (HRR). HIIT training sessions were comprised of ten bouts of alternating 1-min "all-out" efforts and 1-min rest; the balance of time was used to perform light aerobic exercise (<50% HRR). Resistance exercise sessions comprised of the following exercises: leg press, knee flexion, chest press and seated row performed on multistation strength training equipment. Resistance training consisted of 4-week phases that alternated between high-intensity, low-volume sessions (four sets of 5–6 repetitions) and low-intensity, high-volume sessions (four sets of 10–12 repetitions). Participants were also encouraged to perform two additional aerobic exercise sessions at home, and were given heart rate monitors to help ensure intensity targets were achieved.

Postintervention testing

Once the targeted weight loss goal was achieved, energy intake was modified to maintain a stable body weight for 3–4 weeks (1.2 ± 0.3% and 1.1 ± 0.1% change in body weight in the Diet-ONLY and Diet+EX groups, respectively) before the testing procedures performed at baseline were repeated. Participants continued to consume packed-out meals throughout final testing and during the inpatient visit. Participants in the Diet+EX group continued the exercise intervention throughout postintervention testing, with the final exercise sessions performed between 1600 hours and 1700 hours on the day of admission for the inpatient metabolic testing visit and from 1600 hours to 1700 hours the following day during the 24-h blood sampling protocol; these sessions involved 60 min of treadmill exercise at 70–80% HRR.

Sample analyses and bioinformatics

Members of the study team who conducted the sample analyses were blinded to participant group assignment.

Plasma substrate, insulin, C-peptide and cytokine concentrations. Plasma glucose concentration was determined using an automated glucose analyzer (Yellow Springs Instrument). Plasma insulin, C-peptide, HbA1c and lipid profile were determined using an automated chemistry analyzer (Cobas 6000, Roche Diagnostics) in the Washington University Core Laboratory for Clinical Studies. Plasma cytokine concentrations were determined using magnetic bead suspension assays and a Luminex 200 analyzer. Plasma BCAA concentrations were determined by gas chromatography–mass spectrometry (GC–MS) using the EZ:faast Amino Acid Analysis Kit (Phenomenex). Plasma NEFA concentrations were determined using a colorimetric assay (Fujifilm). Plasma glucose tracer-to-tracee ratio (TTR) was determined using GC–MS and heptafluorobutyric (HFB) anhydride to form an HFB derivative of glucose[51].

Skeletal muscle RNA sequencing. Total RNA was isolated from frozen samples of skeletal muscle using QIAzol and an RNeasy Mini Kit in combination with an RNase-free DNase Set (Qiagen)[52]. One sample from the Diet+EX group was excluded because of poor RNA quality. Library preparation of the samples was performed with mRNA reverse transcribed to yield cDNA fragments, which were then sequenced using an Illumina NovaSeq 6000 at the UC San Diego IGM Genomics Center. Expression of individual genes are presented as $\log_2$-transformed counts per million reads (CPM).

Plasma proteomics. Plasma proteomic profiles were determined using the SomaScan Assay v4.0 (SomaLogic), which measures relative abundance of circulating proteins using a process that transforms protein concentrations into a corresponding signature of DNA aptamers, which is quantified on a DNA microarray[53]. Standard samples and negative controls are included with each assay; the resulting raw intensities are normalized and calibrated to control for interplate differences. This analysis identified 4,684 unique proteins that were used to compare the effect of weight loss in the Diet-ONLY and Diet+EX groups.

Fecal 16S amplicon sequencing. DNA extraction from fecal samples, amplification of the V4 region of the 16S rRNA gene, and Illumina amplicon sequencing were performed at the University of California San Diego Center for Microbiome Innovation, according to the Earth Microbiome Project protocols[54]. Sequence data were processed using QIIME 2 (v2019.10)[55]. Paired-end sequences were processed and denoised using the q2-dada2 plug-in. The phylogenetic tree was constructed using fragment insertion by SATé-enabled phylogenetic placement (SEPP) using the q2-fragment-insertion plug-in[56]. Taxonomic assignment of ASVs was performed using a pretrained Naïve Bayes classifier trained on the Greengenes 13_8 database. Alpha diversity was quantified using Chao richness and Shannon entropy on count tables rarefied to 10,000 reads.

Calculations

Plasma concentration AUCs were calculated using the trapezoidal method[57]. ISRs during the OGTT and 24-h study were calculated using a stepwise ISR function to fit C-peptide concentrations to a two-compartment model of C-peptide kinetics (SAAM II v1.2.1, The Epsilon Group) using population-based C-peptide model parameters[58].

Whole-body ICR (that is, the volume of plasma cleared of insulin per minute) over a given time interval during the OGTT and 24-h study was calculated by dividing the ISR by the plasma insulin concentration after adjusting for insulin that accumulated and did not clear from plasma: $ICR_{xy} = [AUC\ ISR_{xy} - V \times (I_y - I_x)]/AUC\ Ins_{xy}$, where subscripts x and y denote time points for the interval, ISR_{xy} is the ISR over the time interval, V is the distribution volume for insulin (assumed to be 55 mL/kg FFM), and I_x and I_y are the plasma insulin concentrations at time points x and y, respectively[59]. HOMA-IR was calculated by dividing the product of the plasma concentrations of insulin (in µU/mL) and glucose (in mmol/L) by 22.5 (ref. 60). The Matsuda insulin sensitivity index was calculated based on plasma glucose (in mg/dL) and insulin (in µU/mL) concentrations measured before and at 30, 60, 90 and 120 min after 75 g glucose ingestion[27]. The hepatic insulin sensitivity index was calculated as the inverse of the product of plasma insulin concentration (in µU/mL) and endogenous glucose rate of appearance (R_a) into the systemic circulation per kilogram FFM. Glucose R_a was determined by dividing the glucose tracer infusion rate by the average plasma glucose TTR during the last 20 min of the basal period of the clamp procedure[26]. Insulin-stimulated glucose rate of disposal (R_d) was assumed to be equal to the sum of the endogenous glucose R_a into the bloodstream and the rate of infused glucose during the last 20 min of the clamp procedure[26]. Whole-body insulin sensitivity was calculated as glucose R_d per kilogram FFM divided by the average plasma insulin concentration in pmol/L (glucose R_d/I) during the final 20 min of the clamp procedure, which primarily represents skeletal muscle insulin sensitivity[21]. β-cell function was assessed as the change in ISR in relation to the change in plasma glucose concentration during the first 30 min of the OGTT, when plasma glucose concentration rapidly increases[61].

Data collection and management

Study data were collected and managed using REDCap v7 electronic data capture tools hosted at Washington University School of Medicine.

Statistical analyses

Primary and secondary cardiometabolic outcomes. The statistical significance of group differences in the intervention-induced changes in primary and secondary outcomes were analyzed using two-tailed analysis of covariance (ANCOVA) with the baseline value as the covariate. In addition, 95% confidence intervals around the mean change within groups are provided to show the range of the change for each variable. Statistical analyses were performed using Excel (v2016, Microsoft) and SPSS (v28, IBM). No P value corrections for multiple testing to control studywide error rate were applied for secondary outomes.

Secondary outcomes related to potential mechanisms involved in regulating insulin sensitivity: muscle transcriptome, plasma proteome and gut microbiome

RNA sequencing. Differential expression analysis was performed comparing before versus after weight loss in each group using the edgeR R/Bioconductor package (v3.40.2). Genes with $|\log_2(FC)| > 0.3$ (where FC is fold change) and two-sided false discovery rate (FDR) < 0.05 were considered DEGs. Functional enrichment analyses were performed on DEGs using DAVID Bioinformatics Resources v6.8 with the GO-BP Direct category. GO terms with two-sided FDR < 0.01 were considered significantly enriched. To confirm the results of the DEG analysis, we identified key genes from the pathway analysis and analyzed these data using ANCOVA.

SomaScan proteomics. Statistical analyses were performed on the hybridization-normalized and median-normalized, plate-scaled protein mean fluorescence intensity (MFI) data after $\log_2$ transformation and removal of proteins with $\log_2(MFI) < 7.5$, leaving 4,684 unique proteins for analysis. Differential abundance of each protein from before to after weight loss in each group was tested using limma (accounting for the mean-variance trend)[62]. P values were adjusted using the Benjamini–Hochberg method, and a $P_{adj} < 0.20$ was considered significant. Gene set enrichment analysis (GSEA) was conducted with the GO-BP and GO-CC categories using the t-statistics from the limma procedure as input. SomaScan data were processed and analyzed in R using the SomaDataIO R package (v5.1.0.9), and the limma (v3.46.0) and fgsea (v1.26.0) Bioconductor packages.

Gut microbiome. For the 16S microbiome data, all analyses were conducted within each group separately to determine group-specific changes during the interventions, and given that virtually no differences between the groups were detected, a pooled analysis was performed to evaluate the effects of diet-induced weight loss on the gut microbiome. Alpha diversity metrics were analyzed using linear mixed-effects models to account for the effect of individual-level variability. Differences in overall microbiome composition between groups and between time points within groups (beta diversity) were assessed by PERMANOVA using distance matrices with the weighted UniFrac distance through Adonis using the following formula: study_group*study_period + sex. Adonis analysis was performed using R (v4.1.3) and Bioconductor (v2.6.2) with the vegan package (v3.14). Differential abundance analysis was performed using a Bayesian linear mixed-effects approach accounting for individual as a random effect. Study group, study period (T0–T3), their interaction and sex were considered as fixed effects. Individual taxa were fit using a negative binomial model accounting for compositionality and overdispersion. For log-ratio analysis, the abundance sum of the ten most associated ASVs was divided by the abundance sum of the ten least associated ASVs (plus pseudocount of 1) in each sample.

Power analysis. We estimated that eight participants per group would be needed to detect a 50% difference in the increase in whole-body (muscle) insulin sensitivity between groups, with a power of 0.9, a two-tailed test, and an α value of 0.05. This power calculation was based on the following assumptions: (1) the variability in muscle insulin sensitivity would be ~10 nmol/kg FFM/min per pmol/L of insulin, which is the variability we have observed previously in a similar study population[63]; and (2) weight loss in the Diet-ONLY group would double muscle insulin sensitivity (estimated to increase from 30 nmol/kg FFM/min per pmol/L of insulin to 60 nmol/kg FFM/min per pmol/L of insulin[2]).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Source data for Figs. 1, 2, 3, 4, 5c and 6 and Extended Data Tables 1, 2, 3 and 4 are provided with this paper. The RNA-seq data generated during this study are available at the National Center for Biotechnology Information (NCBI) Gene Expression Omnibus (GEO) database under accession number GSE230002. We used DAVID Bioinformatics Resources v6.8 (http://david.ncifcrf.gov) to analyze the muscle RNA-seq data. The 16S data have been uploaded to the European Nucleotide Archive (ENA; https://www.ebi.ac.uk/ena) under the study identifier PRJEB61649. The Greengenes 13_8 database was used for taxonomic assignment of the ASVs detected in this study (https://data.qiime2.org/2022.11/common/gg-13-8-99-515-806-nb-classifier.qza). Source data are provided with this paper.

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Author contributions
J.W.B., B.D.K., G.I.S., G.R., J.Y., R.K. and B.W.P. performed sample and data analyses. J.W.B., B.D.K., G.I.S., G.G.S., K.K. and M.L.K. conducted the clinical studies. G.I.S. and S.K. designed the study. J.W.B., B.D.K., G.I.S., B.W.P., R.K. and S.K. interpreted the data and wrote the manuscript. All authors critically reviewed and edited the paper.

Competing interests
S.K. serves on scientific advisory boards for Altimmune and Merck. All other authors declare they have no competing interests.

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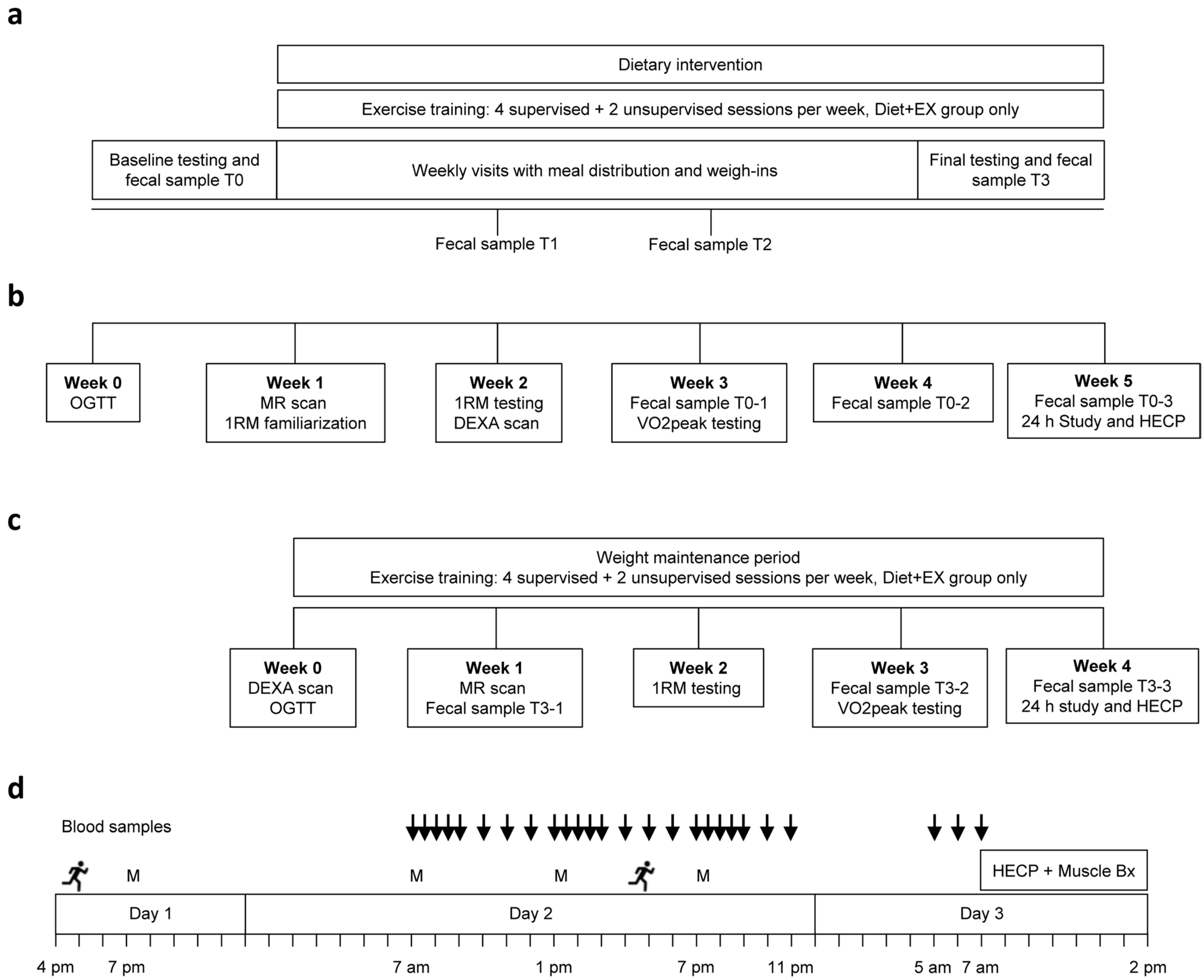

Extended Data Fig. 1 | Schematic diagrams of overall study protocol (a), schedule of individual assessments before (b) and after (c) the intervention and the protocol for inpatient 24-h study blood sampling schedule and hyperinsulinemic-euglycemic clamp procedure (HECP) plus muscle biopsy (Bx) (d). Blood sample timepoints are indicated by arrows and running figure indicates time of a 60-min bout of exercise performed in the Diet+EX group after the intervention. M=meal.

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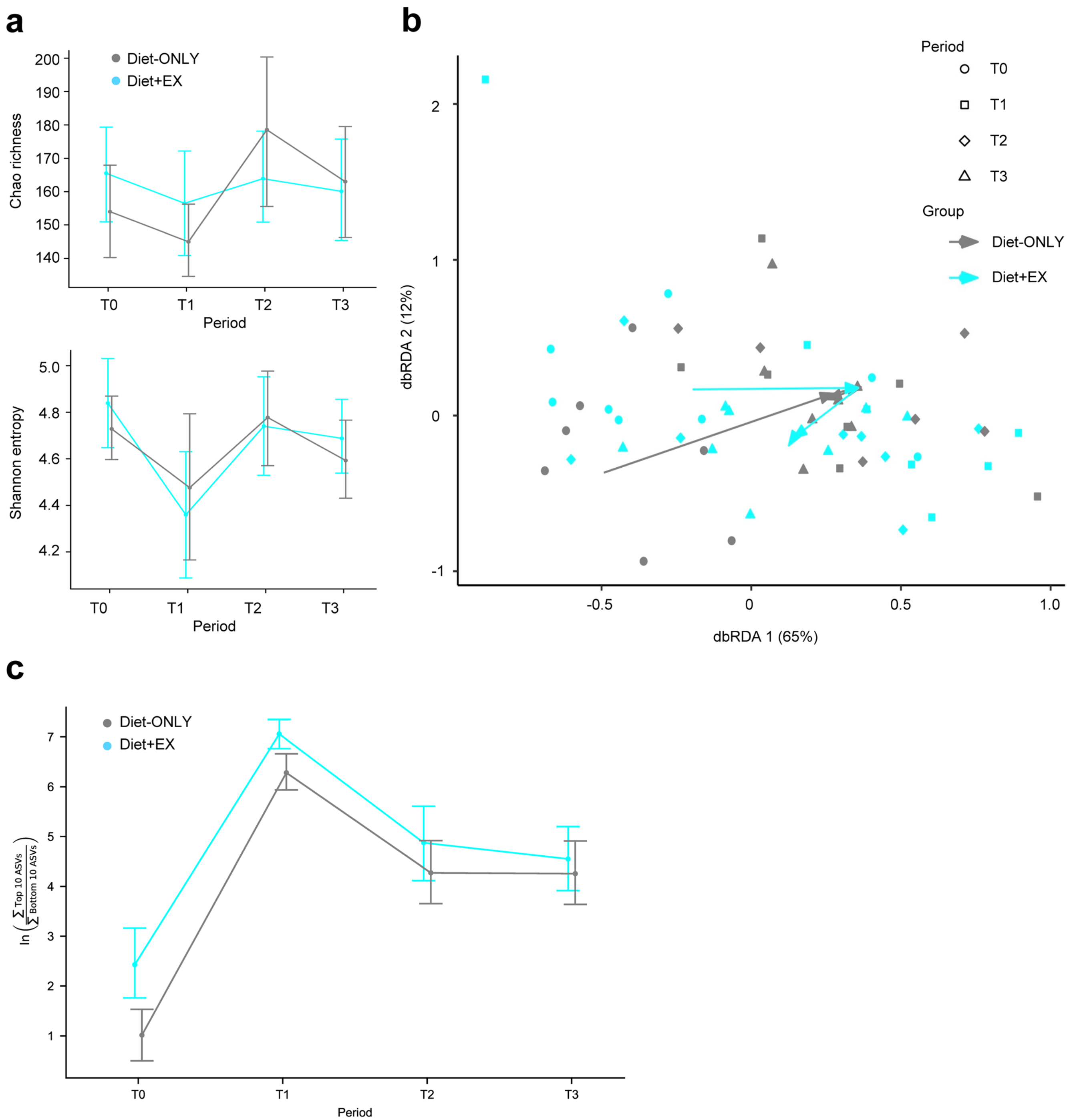

Extended Data Fig. 2 | Effect of each intervention on the gut microbiome. Measures of alpha diversity using both Chao richness (top) and Shannon entropy (bottom), before (T0) and at ~1 month (T1, 5.2 ± 0.7% weight loss), ~2 months (T2, 8.6 ± 0.8% weight loss), and ~4-5 months (T3, 10.5 ± 2.7% weight loss) after starting the intervention in the Diet-ONLY (n = 7) and Diet+EX (n = 8) groups. Data are means ± SEM (**a**). Beta diversity using Distanced-based Redundancy Analysis of the weight UniFrac distance, conditional on the between-person variability (**b**). Log-ratio analysis of 10 ASVs that increased the most and the 10 ASVs that decreased the most from T0 to T1 with treatment groups combined shown over time when stratified by study group. Data are log-ratio means ± SEM (**c**).

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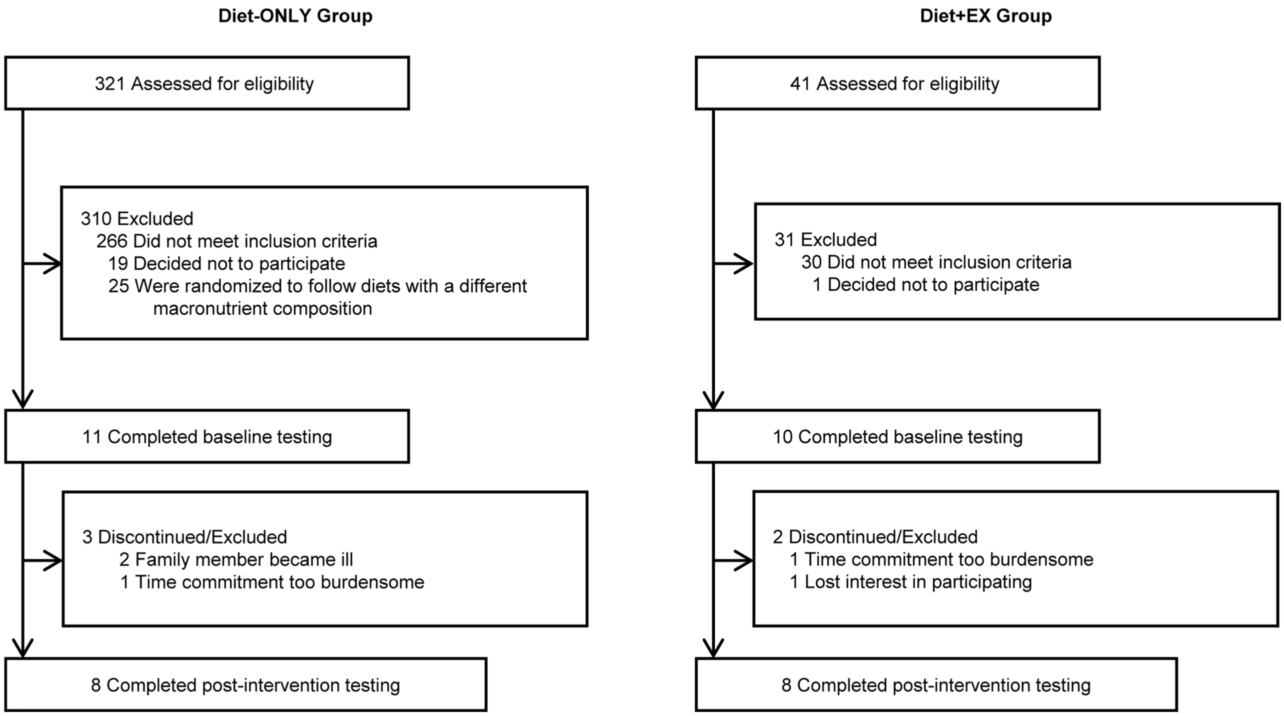

Extended Data Fig. 3 | Flow of study participants.

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Extended Data Table 1 | Strength and cardiovascular fitness

	Diet-ONLY (*n*=6-7)			Diet+EX (*n*=8)			Between Groups
	Before	After	Difference (95% CI)	Before	After	Difference (95% CI)	*P-value*
Leg press (1-RM, kg)[a]	116 ± 24	123 ± 34	7 (-26, 39)	82 ± 7	94 ± 8	13 (2, 23)	0.171
Knee flexion (1-RM, kg)[a]	89 ± 10	85 ± 10	-4 (-11, 2)	65 ± 10	74 ± 9	9 (5, 13)	0.004
Seated row (1-RM, kg)[a]	77 ± 10	75 ± 9	-2 (-3, 0)	69 ± 8	78 ± 10	9 (-1, 18)	0.030
Chest press (1-RM, kg)[a]	69 ± 15	60 ± 12	-8 (-18, 1)	51 ± 7	56 ± 7	4 (-3, 12)	0.068
VO_{2peak} (mL/kg FFM/min)[b]	48.6 ± 4.5	47.2 ± 4.2	-1.4 (-2.7, 0)	48.2 ± 4.9	54.6 ± 6.2	6.4 (1.5, 11.3)	0.005
VO_{2peak} (mL/kg/min)[b]	27.2 ± 3.3	28.7 ± 3.5	1.5 (0.7, 2.4)	25.2 ± 2.5	31.2 ± 3.7	6.0 (2.7, 9.4)	0.004

Values before and after weight loss are expressed as means ± SEM. Abbreviations: 1-RM=one repetition maximum; FFM=fat-free mass; VO_{2peak}=peak oxygen consumption measured by using indirect calorimetry during incremental treadmill exercise to volitional exhaustion. Change values are expressed as mean and 95% confidence bounds within each group. [a] Diet-ONLY group *n*=6. [a] Diet-ONLY group *n*=7. Two-sided analysis of covariance without correction for multiple comparisons was used to determine between group differences after adjusting for values before weight loss.

Values before and after weight loss are expressed as means±SEM. Abbreviations: 1-RM=one repetition maximum; FFM=fat-free mass; VO2peak=peak oxygen consumption measured by using indirect calorimetry during incremental treadmill exercise to volitional exhaustion. Change values are expressed as mean and 95% confidence bounds within each group. a Diet-ONLY group n=6. a Diet-ONLY group n=7. Two-sided analysis of covariance without correction for multiple comparisons was used to determine between group differences after adjusting for values before weight loss.

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Extended Data Table 2 | Fasting plasma markers of inflammation

	Diet-ONLY (*n*=8)			Diet+EX (*n*=8)			Between Groups
	Before	After	Difference (95% CI)	Before	After	Difference (95% CI)	*P-value*
PAI-1 (ng/mL)	63.8 ± 18.5	28.4 ± 6.0	-35.4 (-80.5, 9.8)	34.5 ± 8.2	15.9 ± 3.9	-18.6 (-34.0, -3.1)	0.640
IFN-γ (pg/mL)	15.2 ± 3.2	15.6 ± 3.4	0.4 (-1.3, 2.1)	11.4 ± 1.7	10.7 ± 1.9	-0.7 (-2.3, 0.8)	0.404
IL-1β (pg/mL)	0.75 ± 0.09	0.76 ± 0.09	0.01 (-0.21, 0.23)	0.69 ± 0.09	0.69 ± 0.11	-0.00 (-0.08, 0.08)	0.853
IL-6 (pg/mL)	18.6 ± 9.9	12.9 ± 8.0	-5.7 (-15.7, 4.3)	16.1 ± 10.1	12.1 ± 7.8	-4.0 (-9.7, 1.7)	0.550
IL-10 (pg/mL)	15.6 ± 3.8	13.9 ± 3.6	-1.7 (-5.9, 2.4)	18.9 ± 7.5	16.1 ± 6.3	-2.8 (-7.1, 1.4)	0.459
IL-17 (pg/mL)	5.9 ± 1.9	6.2 ± 1.9	0.3 (-0.8, 1.5)	3.4 ± 0.9	3.0 ± 0.9	-0.4 (-1.2, 0.4)	0.214
MCP-1 (pg/mL)	8.8 ± 0.4	8.2 ± 0.5	-0.6 (-1.1, 0.0)	8.6 ± 0.6	8.1 ± 0.6	-0.5 (-1.2, 0.2)	0.879
TNF-α (pg/mL)	4.8 ± 0.4	5.0 ± 0.4	0.2 (-0.4, 0.7)	5.0 ± 0.5	4.7 ± 0.6	-0.3 (-1.1, 0.4)	0.191
CRP (mg/L)	5.7 ± 1.2	5.3 ± 1.2	-0.4 (-1.3, 0.5)	5.4 ± 1.2	4.5 ± 1.1	-0.9 (-3.3, 1.6)	0.615

Values before and after weight loss are expressed as means ± SEM. Change values are expressed as mean and 95% confidence bounds within each group. Two-sided analysis of covariance without correction for multiple comparisons was used to determine between group differences after adjusting for values before weight loss.

Values before and after weight loss are expressed as means±SEM. Change values are expressed as mean and 95% confidence bounds within each group. Two-sided analysis of covariance without correction for multiple comparisons was used to determine between group differences after adjusting for values before weight loss.

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Extended Data Table 3 | Metabolic outcomes before and after intervention

	Diet-ONLY (*n*=8)			Diet+EX (*n*=8)			Difference between Groups
	Before	After	Difference (95% CI)	Before	After	Difference (95% CI)	*P*-value
3-h oral glucose tolerance test[a]							
Glucose AUC (mg/dL x 3h)	476 ± 27	418 ± 11	-59 (-118, 1)	451 ± 8	404 ± 17	-47 (-77, -17)	0.790
Insulin AUC ([pmol/L x 3h] x 10^3)	2.91 ± 0.39	2.07 ± 0.20	-0.83 (-1.54, -0.13)	2.65 ± 0.44	1.37 ± 0.18	-1.29 (-2.10, -0.48)	0.008
ISR AUC ([pmol/min x 3h] x 10^3)	4.52 ± 0.52	3.64 ± 0.40	-0.89 (-1.51, -0.26)	3.61 ± 0.31	2.88 ± 0.16	-0.73 (-1.29, -0.18)	0.394
ICR AUC (L/min x 3h)	5.42 ± 0.38	5.80 ± 0.37	0.38 (-0.45, 1.22)	5.14 ± 0.69	7.49 ± 1.00	2.35 (0.31, 4.39)	0.082
24-h blood sampling study[b]							
Glucose AUC ([mg/dL x 24h] x 10^3)	2.66 ± 0.09	2.45 ± 0.06	-0.22 (-0.37, -0.06)	2.59 ± 0.03	2.36 ± 0.03	-0.23 (-0.35, -0.11)	0.334
Insulin AUC ([pmol/L x 24h] x 10^3)	11.3 ± 1.3	9.0 ± 1.5	-2.4 (-4.1, -0.6)	13.2 ± 3.0	6.0 ± 0.8	-7.2 (-13.5, -0.9)	0.030
ISR AUC ([pmol/min x 24h] x 10^3)	19.7 ± 2.1	15.3 ± 2.3	-4.5 (-7.9, -1.0)	17.3 ± 2.2	12.2 ± 1.1	-5.1 (-8.6, -1.6)	0.460
ICR AUC (L/min x 24h)	47.6 ± 2.6	51.4 ± 6.5	3.8 (-7.4, 14.9)	42.1 ± 4.6	61.4 ± 5.8	19.2 (12.9, 25.6)	0.009
NEFA AUC (mmol/L x 24h)	7.50 ± 0.70	5.92 ± 0.46	-1.57 (-2.95, -0.20)	6.27 ± 0.54	8.08 ± 0.39	1.81 (0.00, 3.61)	0.003
Insulin sensitivity							
Whole-body (muscle) insulin sensitivity	32.7 ± 3.3	48.4 ± 7.2	15.7 (4.7, 26.7)	38.4 ± 6.7	80.6 ± 7.6	42.2 (27.6, 56.8)	0.006
Hepatic insulin sensitivity index	2.72 ± 0.43	5.11 ± 0.89	2.39 (1.27, 3.51)	2.92 ± 0.33	7.67 ± 0.94	4.75 (2.95, 6.55)	0.014
Matsuda index[a]	1.39 ± 0.20	1.83 ± 0.16	0.44 (0.00, 0.88)	1.63 ± 0.44	2.91 ± 0.38	1.27 (0.18, 2.36)	0.037

Values before and after weight loss are expressed as means ± SEM. Abbreviations: ICR=insulin clearance rate; ISR=insulin secretion rate; NEFA=non-esterified fatty acids. Change values are expressed as mean and 95% confidence bounds within each group. a Diet-ONLY group n=7. b Diet+EX group n=7. Whole-body insulin sensitivity represents glucose disposal rate from plasma per kilogram fat-free mass (FFM) divided by plasma insulin concentration during the hyperinsulinemic-euglycemic clamp procedure, which primarily represents skeletal muscle insulin sensitivity. Hepatic insulin sensitivity index was calculated as the reciprocal of the product of endogenous glucose production rate per kilogram FFM and plasma insulin concentration during basal conditions. The Matsuda insulin sensitivity index was calculated as 10,000 divided by the square root of the product of plasma glucose and insulin concentrations during the basal state and mean plasma glucose and insulin concentrations at 30, 60, 90 and 120 min during the oral glucose tolerance test. Two-sided analysis of covariance without correction for multiple comparisons was used to determine between group differences after adjusting for values before weight loss.

Values before and after weight loss are expressed as means ± SEM. Abbreviations: ICR=insulin clearance rate; ISR=insulin secretion rate; NEFA=non-esterified fatty acids. Change values are expressed as mean and 95% confidence bounds within each group. a Diet-ONLY group n=7. b Diet+EX group n=7. Whole-body insulin sensitivity represents glucose disposal rate from plasma per kilogram fat-free mass (FFM) divided by plasma insulin concentration during the hyperinsulinemic-euglycemic clamp procedure, which primarily represents skeletal muscle insulin sensitivity. Hepatic insulin sensitivity index was calculated as the reciprocal of the product of endogenous glucose production rate per kilogram FFM and plasma insulin concentration during basal conditions. The Matsuda insulin sensitivity index was calculated as 10,000 divided by the square root of the product of plasma glucose and insulin concentrations during the basal state and mean plasma glucose and insulin concentrations at 30, 60, 90 and 120 min during the oral glucose tolerance test. Two-sided analysis of covariance without correction for multiple comparisons was used to determine between group differences after adjusting for values before weight loss.

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Extended Data Table 4 | Expression of key genes related to mitochondrial biogenesis, energy metabolism and angiogenesis in skeletal muscle

	Diet-ONLY (*n*=6)			Diet+EX (*n*=7)			Difference between Groups
	Before	After	Difference (95% CI)	Before	After	Difference (95% CI)	*P*-value
PPARGC1A	6.04 ± 0.18	6.16 ± 0.17	0.12 (-0.28, 0.52)	6.07 ± 0.12	6.84 ± 0.10	0.77 (0.57, 0.97)	0.001
MFF	5.06 ± 0.03	4.96 ± 0.09	-0.10 (-0.34, 0.14)	4.97 ± 0.06	5.26 ± 0.03	0.29 (0.17, 0.40)	0.009
CS	8.00 ± 0.09	8.04 ± 0.06	0.04 (-0.06, 0.13)	7.97 ± 0.04	8.37 ± 0.06	0.40 (0.24, 0.55)	<0.001
MDH1	7.26 ± 0.12	7.24 ± 0.17	-0.02 (-0.32, 0.29)	7.15 ± 0.08	7.80 ± 0.15	0.65 (0.33, 0.98)	0.007
CYCS	6.33 ± 0.13	6.36 ± 0.17	0.03 (-0.35, 0.42)	6.43 ± 0.09	7.11 ± 0.11	0.67 (0.45, 0.90)	0.003
ATP5F1A	9.05 ± 0.10	9.02 ± 0.12	-0.03 (-0.24, 0.18)	8.81 ± 0.09	9.27 ± 0.10	0.47 (0.19, 0.74)	0.030
ACADVL	8.96 ± 0.04	8.72 ± 0.10	-0.24 (-0.51, 0.02)	8.66 ± 0.08	9.04 ± 0.12	0.38 (0.14, 0.62)	0.024
ACADM	6.85 ± 0.13	6.54 ± 0.21	-0.31 (-0.68, 0.06)	6.87 ± 0.15	7.21 ± 0.14	0.34 (0.22, 0.46)	0.001
LPL	5.20 ± 0.23	5.26 ± 0.18	0.06 (-0.32, 0.44)	5.28 ± 0.18	6.38 ± 0.23	1.11 (0.31, 1.91)	0.006
CD36	8.62 ± 0.12	8.38 ± 0.11	-0.23 (-0.52, 0.05)	8.28 ± 0.12	8.91 ± 0.18	0.63 (0.38, 0.89)	0.001
VEGFA	6.63 ± 0.14	6.48 ± 0.18	-0.15 (-0.69, 0.38)	6.59 ± 0.08	7.26 ± 0.13	0.67 (0.42, 0.92)	0.004
KDR	3.38 ± 0.13	3.20 ± 0.06	-0.18 (-0.52, 0.16)	3.19 ± 0.09	4.08 ± 0.12	0.89 (0.58, 1.19)	<0.001

Values before and after weight loss are log_2CPM and expressed as means ± SEM. Change values are expressed as mean and 95% confidence bounds within each group. Two-sided analysis of covariance without correction for multiple comparisons was used to determine between group differences after adjusting for values before weight loss.

Expression of key genes related to mitochondrial biogenesis, energy metabolism and angiogenesis in skeletal muscle.

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Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

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☐	☒	The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement
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☐	☒	For null hypothesis testing, the test statistic (e.g. F, t, r) with confidence intervals, effect sizes, degrees of freedom and P value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐	Estimates of effect sizes (e.g. Cohen's d, Pearson's r), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	Data was collected in Redcap version 7
Data analysis	Data were analyzed using SPSS version 28 and microsoft excel version 16 C-peptide kinetics were analyzed by using SAAM II software version 1.2.1, The Epsilon Group, Charlottesville, VA Differential gene expression analysis was performed by using the edgeR R/Bioconductor package (version 3.40.2) SomaScan proteomics data were processed and analyzed in R using the SomaDataIO R package (version 5.1.0.9), and the limma (3.46.0), and fgsea (1.26.0) Bioconductor packages Gut microbiome sequence data were processed by using QIIME2 (v2019.10) and analyzed by using R (Version 4.1.3, The R Foundation for Statistical Computing, Vienna, Austria) and Bioconductor (Version 2.6.2) with the vegan package (Version 3.14).

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

Source data for Table 1, Figures 1, 2, 3, 4, 5c, and 6 and Extended Data Tables 1, 2, 3 and 5 are provided with this paper. The RNA-seq data generated during this study is available at the NCBI GEO database (accession no. GSE230002). We used the Database for Annotation, Visualization, and Integrated Discovery (DAVID) Bioinformatics Resources 6.8 to analyze the muscle RNA-seq data and it is available at http://david.ncifcrf.gov/. The 16S data has been uploaded through EBI https://www.ebi.ac.uk/ena under the study identifier PRJEB61649 and the GreenGenes 13_8 database was used for taxonomic assignment for the ASVs detected in this study (https://data.qiime2.org/2022.11/common/gg-13-8-99-515-806-nb-classifier.qza).

Human research participants

Policy information about studies involving human research participants and Sex and Gender in Research.

Reporting on sex and gender	Only self-reported sex was included. Findings apply to both sexes equally.
Population characteristics	Shown in Table 1 in the manuscript and in the first paragraph on the Methods
Recruitment	All participants were recruited by using the Volunteers for Health database at Washington University School of Medicine and by local postings between April 2016 and November 2018. All participants expressed willingness to follow study protocols to achieve weight loss, which may suggest the potential for self-selection bias. However, this was true for both study groups and therefore is unlikely to pose a significant contributor to our findings.
Ethics oversight	Institutional Review Board within the Human Research Protection Office at Washington University School of Medicine

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Power analysis. We estimated that 8 participants/group would be needed to detect a 50% difference in the increase in whole-body (muscle) insulin sensitivity between groups, with a power of 0.9, a two-tailed test, and an α-value of 0.05. This power calculation was based on the following assumptions: i) the variability in muscle insulin sensitivity would be ~10 nmol/kg FFM/min per pmol/L of insulin, which is the variability we have observed previously in a similar study population; and ii) weight loss in the Diet-ONLY group would double muscle insulin sensitivity (estimated to increase from 30 nmol/kg FFM/min per pmol/L of insulin to 60 nmol/kg FFM/min per pmol/L of insulin), as we have found previously.
Data exclusions	No data were excluded from analyses
Replication	Replication experiments were not conducted due to participant and resource availability.
Randomization	Eligible participants were enrolled in either NCT02706262 and NCT02706288. Randomization was not possible, however, inclusion criteria for the two trials were identical resulting in no differences in subject characteristics at baseline assessed by Students t-tests.
Blinding	Participants in the trials were not blinded due to the nature of diet and exercise interventions, which are impossible to obscure from the participant. Sample analyses were performed by members of the study team who were blinded to group assignment.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☐	☒ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Clinical data

Policy information about clinical studies

All manuscripts should comply with the ICMJE guidelines for publication of clinical research and a completed CONSORT checklist must be included with all submissions.

Clinical trial registration

NCT02706262 and NCT02706288

Study protocol

Protocols are included with submission

Data collection

Study visits were conducted in the Clinical Translational Research Unit (CTRU) and the Center for Clinical Imaging Research at Washington University School of Medicine in St. Louis, MO between April 2016 and November 2018

Outcomes

Our primary outcome was whole-body (primarily muscle) insulin sensitivity assessed by using the hyperinsulinemic-euglycemic clamp procedure in conjunction with stable isotopically labelled glucose tracer infusion
In addition, we assessed several secondary outcomes including:
1) liver insulin sensitivity assessed by using the hyperinsulinemic-euglycemic clamp procedure in conjunction with stable isotopically labelled glucose tracer infusion
2) metabolic response to glucose ingestion assessed for 3 hours and ingestion of multiple mixed-meals assessed over 24 hours
3) β-cell function and insulin kinetics assessed by modeling of the C-peptide and insulin data after glucose ingestion assessed for 3 hours and ingestion of multiple mixed-meals assessed over 24 hours.
4) Fasting plasma lipid profile, markers of inflammation, plasma branched-chain amino acid (BCAA) concentrations,
5) body composition (fat-free mass, fat mass and intrahepatic triglyceride [IHTG] content) assessed by using dual energy X-ray absorptiometry and magnetic resonance imaging,
6) cardiorespiratory fitness during a maximal treadmill test to exhaustion
7) muscle strength
8) skeletal muscle tissue global transcriptomics assessed using RNA sequencing
9) the fasting plasma proteome using the SOMAscan proteomics assay
10) the composition of the gut microbiome assessed by using 16s rRNA amplicon sequencing